

Probability, Reliability and Quality

Part A: Inspections

Name: Zhifeng Song

Summary

This study conducted a measurement system assessment for 3D-printed hinges. A design of 10 pieces $\times$ 3 operators $\times$ 3 repetitions (90 trials) was employed, and Minitab was used to analyse three KPIs—thickness, width, and length—using a cross-type Gage R&R (ANOVA) analysis. Taking thickness as an example: %GRR = 9.81%, ndc = 14; the effects of operators and their interactions were not significant ($p > 0.3$), and the total variation primarily originated from part-to-part differences (Part-to-Part $\approx$ 99%), indicating that the gage capability is acceptable and stable. It is recommended to standardise the reference points and measurement postures, strengthen calibration and documentation, and, if necessary, configure positioning fixtures and higher-resolution gauges to support future KPI monitoring.

Introduction、



Figure1 QI reliability test fixture

Before discussing manufacturing capabilities, it is essential to validate the reliability of the measurement system. This report uses the 3D-printed hinge provided in the course as the subject, focusing on three Key Performance Indicators (KPIs): thickness, width, and length. The Minitab Cross-Type Gage R&R (ANOVA) method is employed to decompose measurement variation (repeatability, reproducibility, and between-part variation). Using %Study Var (%GRR) and ndc as criteria, the objectives are: ① to determine whether the gage meets the 'acceptable' threshold; ② to identify the primary sources of error; ③ to establish actionable measurement specifications and improvement recommendations. Document structure: Part 1 records printing and equipment; Part 2 provides part identification and KPI definitions; Part 3 presents the MSA process, charts, and conclusions.

3D Printing Process

Step	Process
1	CAD Design
2	File Conversion
3	Slicing
4	Setting 3D Printer
5	Printer Process
6	Post Processing
7	Check Product
8	KPI Measurement

KPI Analysis and Measurement Evaluation

Definition KPIS

Thickness: As shown in the figure, the vertical distance from the bottom of the hinge to the bottom of the groove.

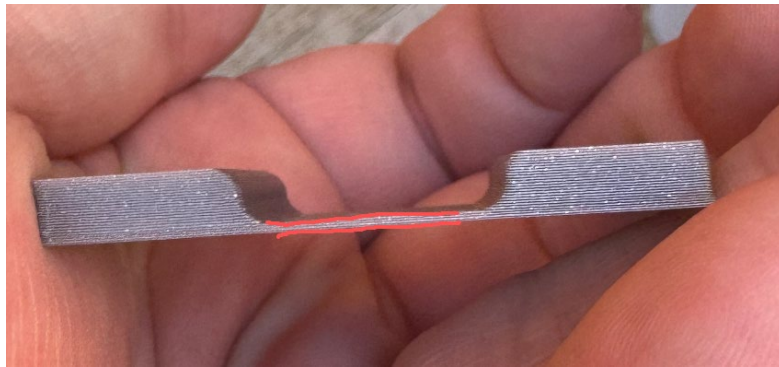


Figure2 Thickness

Length: The length of the internal opening of the hinge, measured in a straight line from one side to the other, as shown in the figure.

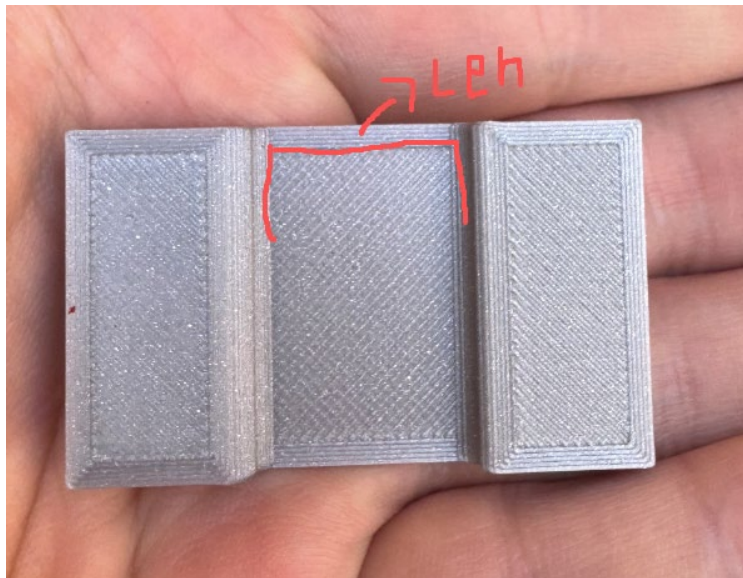


Figure 3 Length

Width: The width of the bottom part of the hinge, measured in a straight line from one edge to the other, as shown in the figure.

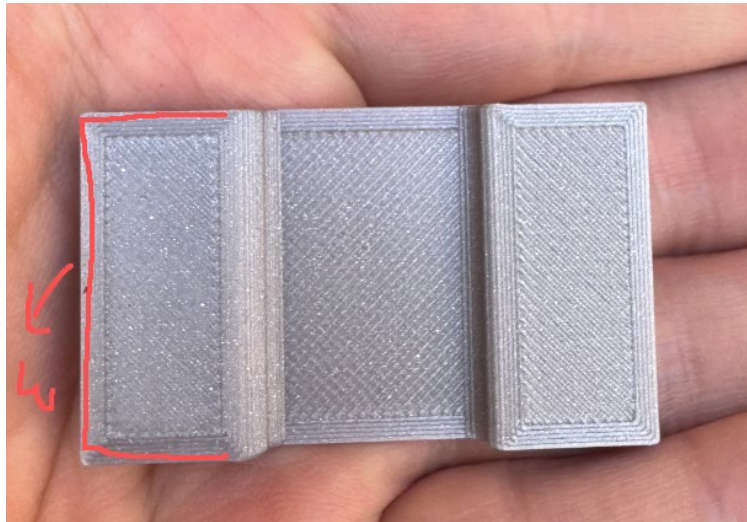


Figure 4 Width

KPI	Definition (baseline and measurement points)	Measurement methods	Measuring instruments and resolution	unit
Thickness	The local plate thickness of the blade is measured once at a representative cross-section in the middle after positioning on the fixture reference surface.	Contact-type two-sided measurement, with the probe perpendicular to the surface and gently closed.	Use a micrometer with a precision of 0.001 mm as the preferred tool. If using a caliper, use one with a precision of 0.01 mm and ensure that the same caliper is used throughout the measurement process.	mm
Width	Nominal width from the outer edge of a single blade to	Flat bonding reference, caliper outer measuring	Digital caliper 0.01 mm	mm

	the centre line	jaws contact both edges		
Length	Total length corresponding to the nominal length/hole spacing along the hinge axis	End face-to-end face (or hole-to-hole) external measurement method, ensuring that the two end faces are in contact with the measuring jaws.	Digital caliper 0.01 mm	mm

Measurement

Thickness: Measure at the mid-section using the fixture surface as the reference plane. Use a micrometre or calliper to gently touch the surface and read the measurement using a ratchet mechanism. Repeat this process multiple times to obtain multiple readings.

Length: Align the two end faces along the axis so that they are fully in contact with the outer measuring jaws of the digital calliper, ensuring they remain in line. Gently press to stabilise the reading, then repeat multiple times to obtain the value.

Width: Under fixture positioning at the mid-section, the digital calliper's outer measuring jaws simultaneously contact both outer edges, parallel to the edge and perpendicular to the thickness direction. Gently press to read the value and repeat multiple times to obtain the measurement.

Summary of KPI measurement data

Thickness	Msmt1	Msmt2	Msmt3	Msmt4	Msmt5	Msmt6	Msmt7	Msmt8	Msmt9
1	0.798	0.79	0.796	0.799	0.8	0.798	0.805	0.785	0.799
2	0.739	0.742	0.732	0.737	0.738	0.721	0.725	0.735	0.736
3	0.856	0.855	0.851	0.859	0.855	0.848	0.856	0.861	0.853

4	0.758	0.759	0.751	0.758	0.752	0.748	0.754	0.758	0.743
5	0.803	0.8	0.804	0.817	0.811	0.792	0.81	0.801	0.803
6	0.72	0.724	0.732	0.731	0.717	0.713	0.733	0.733	0.717
7	0.876	0.868	0.876	0.879	0.869	0.869	0.872	0.872	0.878
8	0.673	0.665	0.666	0.679	0.674	0.665	0.671	0.685	0.682
9	0.715	0.708	0.699	0.711	0.697	0.697	0.712	0.711	0.713
10	0.754	0.744	0.758	0.762	0.753	0.748	0.75	0.752	0.757
Length	Msmt1	Msmt2	Msmt3	Msmt4	Msmt5	Msmt6	Msmt7	Msmt8	Msmt9
1	17.01	16.75	17.45	17.07	17.14	17.7	17.35	17.53	17.67
2	17.28	17.46	17.42	17.81	17.62	17.75	17.67	17.79	17.85
3	17.91	17.17	17.52	17.94	17.91	17.92	17.43	17.51	17.79
4	18.04	18.09	18.99	17.49	17.62	17.6	18.18	17.79	18.77
5	17.59	17.61	18.15	17.54	17.2	17.18	17.68	17.15	18.04
6	18.04	17.35	17.45	17.63	17.54	17.39	17.12	17.83	17.98
7	17.78	17.95	17.55	17.57	17.8	17.47	17.29	17.58	17.17
8	17.78	17.5	17.4	17.82	18.1	17.96	17.53	17.83	18.02
9	17.98	18.4	17.97	17.6	17.49	17.16	17.98	17.93	18.08
10	17.47	17.98	17.73	17.35	17.56	17.65	17.41	17.33	17.96
Width	Msmt1	Msmt2	Msmt3	Msmt4	Msmt5	Msmt6	Msmt7	Msmt8	Msmt9
1	29.93	29.8	29.85	30.09	29.87	29.86	29.86	29.85	29.84
2	29.88	29.92	29.87	30.06	29.89	29.81	29.85	29.82	29.88
3	29.9	29.81	29.88	29.86	29.91	30	29.91	29.86	29.92
4	29.92	29.87	29.94	30.15	29.83	30.05	29.93	29.86	29.91
5	29.84	29.83	29.87	30.03	29.9	29.94	29.85	29.83	29.89
6	30.05	30.05	30.1	30.16	30.03	30.09	30.01	30.04	30.09
7	29.91	29.85	29.88	29.93	29.89	29.9	29.94	29.81	29.9
8	30.36	30.24	30.21	30.3	30.25	30.34	30.4	30.3	30.17

9	29.91	29.87	29.98	29.94	29.95	29.95	29.99	29.98	29.99
10	29.88	29.96	29.89	29.9	29.98	29.9	29.91	29.9	29.9

Overall, the KPI measurement system meets requirements—taking thickness as an example ($\%GRR = 9.81\%$, $ndc = 14$), the total variation mainly comes from differences between parts, and consistency between operators is good.

Gage R&R Study (ANOVA)

Thickness: Variation primarily stems from between parts (approximately 99.0%), with operator ($p=0.301$) and interaction ($p=0.794$) both insignificant; overall %GRR=9.81%, $ndc=14$, the measuring system is acceptable and demonstrates good consistency.

Two-Way ANOVA Table With Interaction

Source	DF	SS	MS	F	P
Part	9	0.332581	0.0369535	1229.51	0.000
Operator	2	0.000077	0.0000386	1.28	0.301
Part * Operator	18	0.000541	0.0000301	0.70	0.794
Repeatability	60	0.002565	0.0000427		
Total	89	0.335764			

α to remove interaction term = 0.05

Figure 5 Two-Way ANOVA For Thickness

Gage R&R

Variance Components

Source	VarComp	%Contribution (of VarComp)
Total Gage R&R	0.0000398	0.96
Repeatability	0.0000398	0.96
Reproducibility	0.0000000	0.00
Operator	0.0000000	0.00
Part-To-Part	0.0041015	99.04
Total Variation	0.0041413	100.00

Process tolerance = 0.2

Gage Evaluation

Source	StdDev (SD)	Study Var (6 × SD)	%Study Var (%SV)	%Tolerance (SV/Toler)
Total Gage R&R	0.0063100	0.037860	9.81	18.93
Repeatability	0.0063100	0.037860	9.81	18.93
Reproducibility	0.0000000	0.000000	0.00	0.00
Operator	0.0000000	0.000000	0.00	0.00
Part-To-Part	0.0640431	0.384259	99.52	192.13
Total Variation	0.0643532	0.386119	100.00	193.06

Number of Distinct Categories = 14

Figure 6 Variation and Evaluation for thickness

Gage R&R (ANOVA) for Thickness

Gage name:
Date of study:

Reported by:
Tolerance:
Misc:

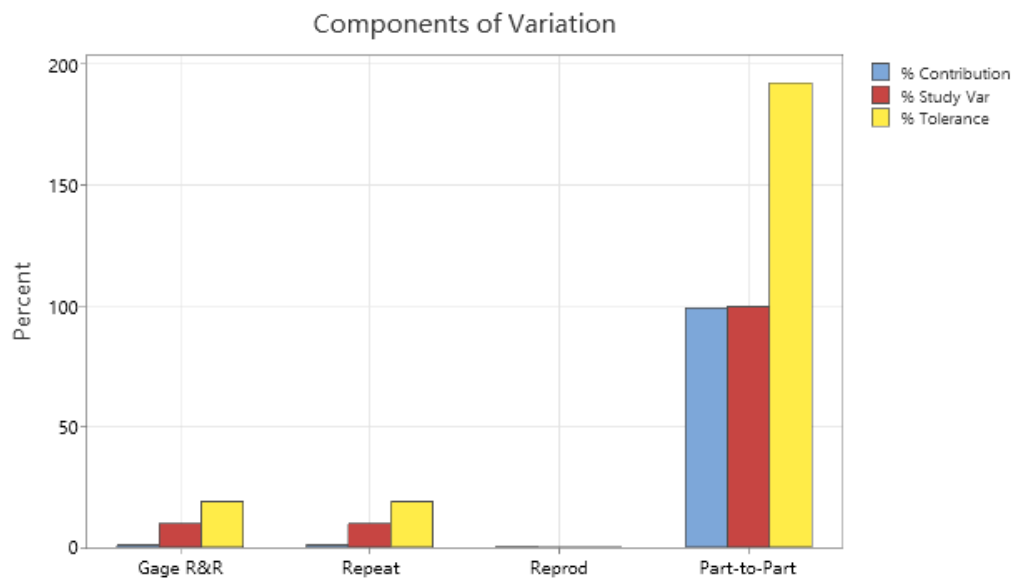


Figure7 Report for thickness

Gage R&R (ANOVA) for Thickness

Gage name:
Date of study:

Reported by:
Tolerance:
Misc:

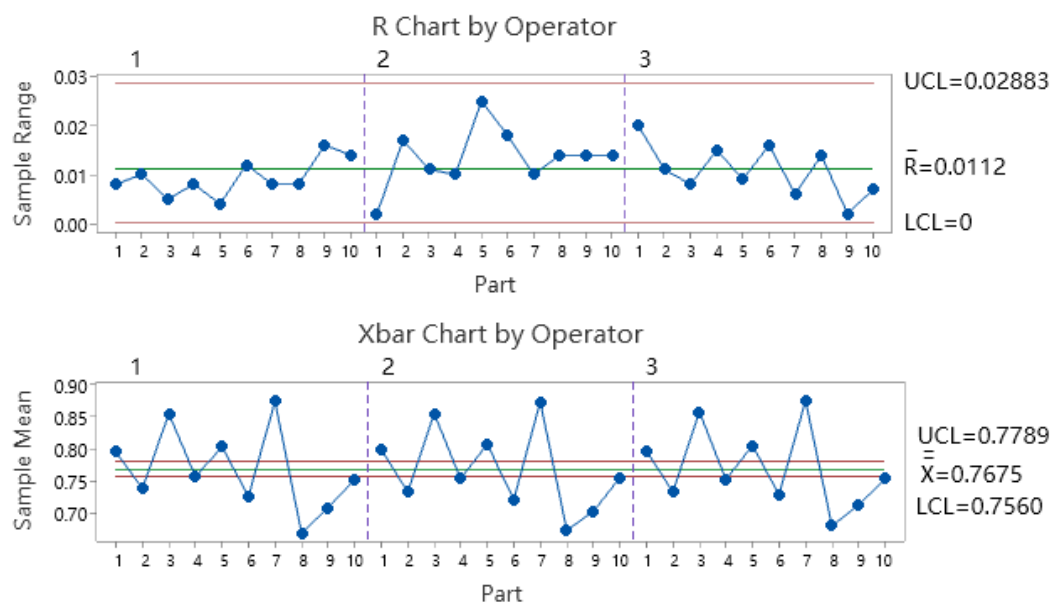


Figure 7 Report for thickness

Gage R&R (ANOVA) for Thickness

Gage name:
Date of study:

Reported by:
Tolerance:
Misc:

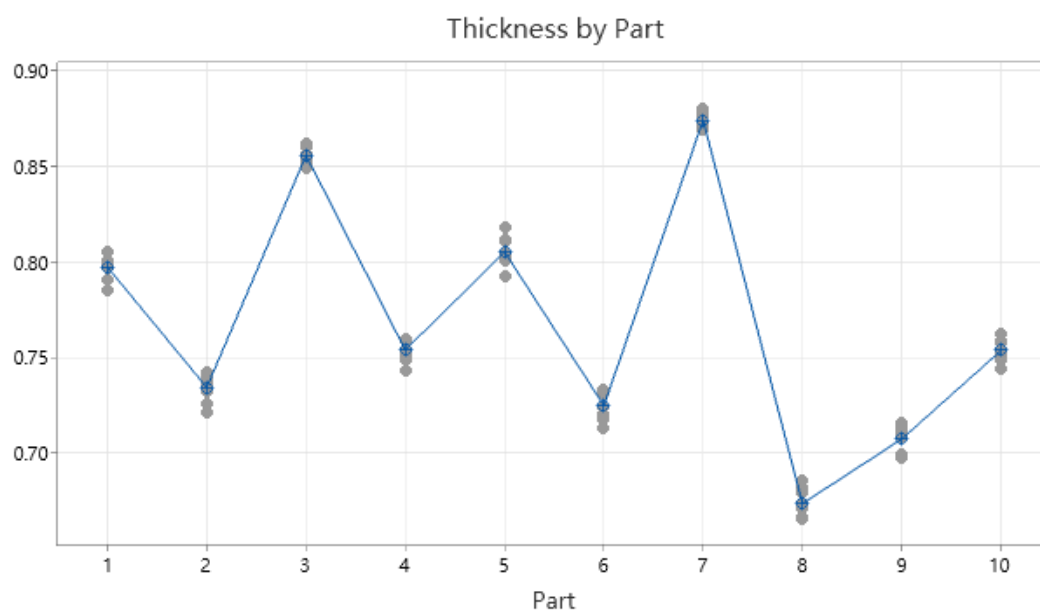


Figure7 Report for thickness

Gage R&R (ANOVA) for Thickness

Gage name:
Date of study:

Reported by:
Tolerance:
Misc:

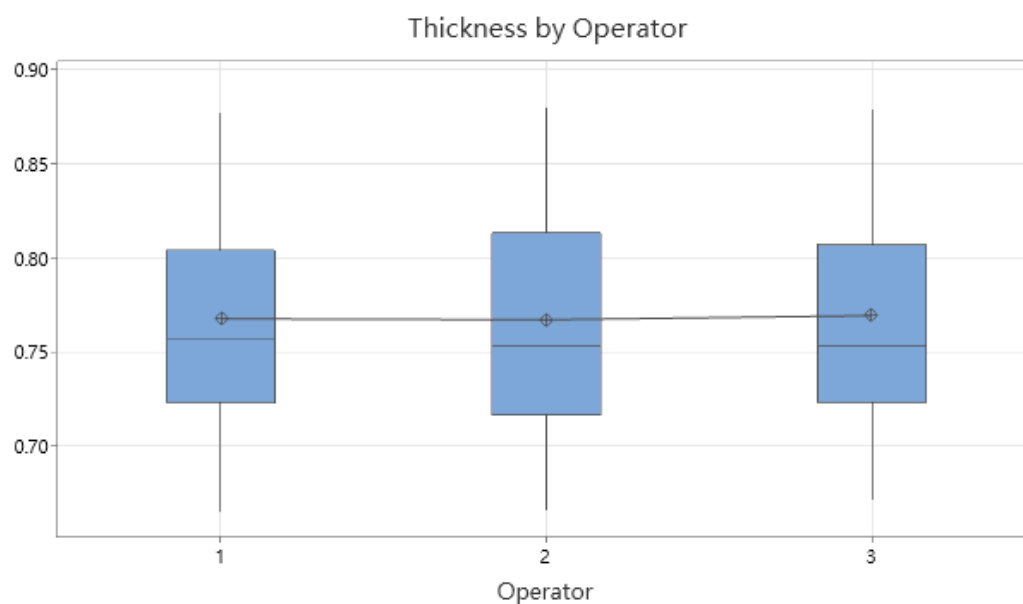


Figure7 Report for thickness

Gage R&R (ANOVA) for Thickness

Gage name:
Date of study:

Reported by:
Tolerance:
Misc:

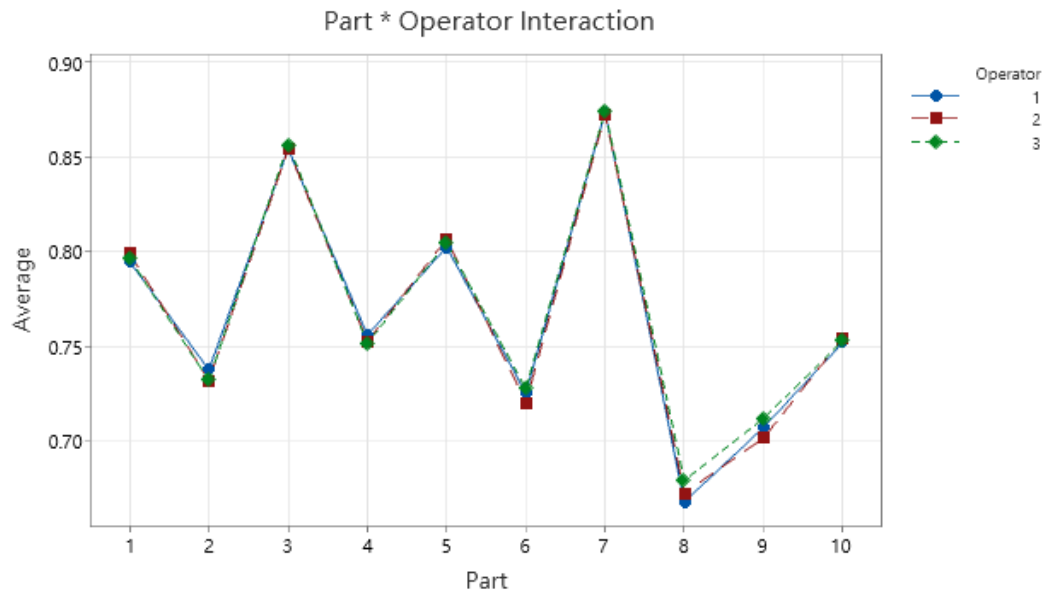


Figure 7 Report for thickness

Length: Unacceptable (%GRR = 92.63%, ndc = 1); Part/Operator not significant, interaction significant ($p = 0.003$); variation primarily due to measuring tools (GRR 85.8% = EV 56.4% + interaction 29.4%, PV 14.2%); R near UCL, Xbar not parallel $\Rightarrow$ poor repeatability and interaction present.

Two-Way ANOVA Table With Interaction

Source	DF	SS	MS	F	P
Part	9	3.2854	0.365048	1.88394	0.121
Operator	2	0.2644	0.132181	0.68216	0.518
Part * Operator	18	3.4878	0.193769	2.56214	0.003
Repeatability	60	4.5377	0.075628		
Total	89	11.5753			

α to remove interaction term = 0.05

Figure8 Two-Way ANOVA For Length

Gage R&R

Variance Components

Source	VarComp	%Contribution (of VarComp)
Total Gage R&R	0.115008	85.80
Repeatability	0.075628	56.42
Reproducibility	0.039380	29.38
Operator	0.000000	0.00
Operator*Part	0.039380	29.38
Part-To-Part	0.019031	14.20
Total Variation	0.134039	100.00

Process tolerance = 1

Gage Evaluation

Source	StdDev (SD)	Study Var (6 × SD)	%Study Var (%SV)	%Tolerance (SV/Toler)
Total Gage R&R	0.339128	2.03477	92.63	203.48
Repeatability	0.275005	1.65003	75.11	165.00
Reproducibility	0.198445	1.19067	54.20	119.07
Operator	0.000000	0.00000	0.00	0.00
Operator*Part	0.198445	1.19067	54.20	119.07
Part-To-Part	0.137953	0.82772	37.68	82.77
Total Variation	0.366114	2.19668	100.00	219.67

Number of Distinct Categories = 1

Figure 9 Variation and Evaluation for Length

Gage R&R (ANOVA) for Length

Gage name:
Date of study:

Reported by:
Tolerance:
Misc:

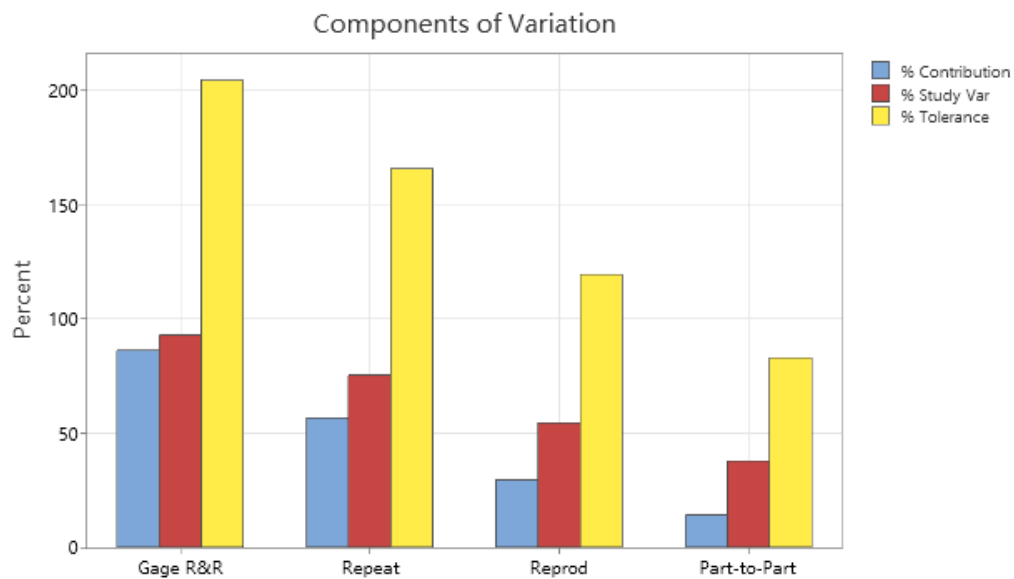


Figure10 Report for Length

Gage R&R (ANOVA) for Length

Gage name:
Date of study:

Reported by:
Tolerance:
Misc:

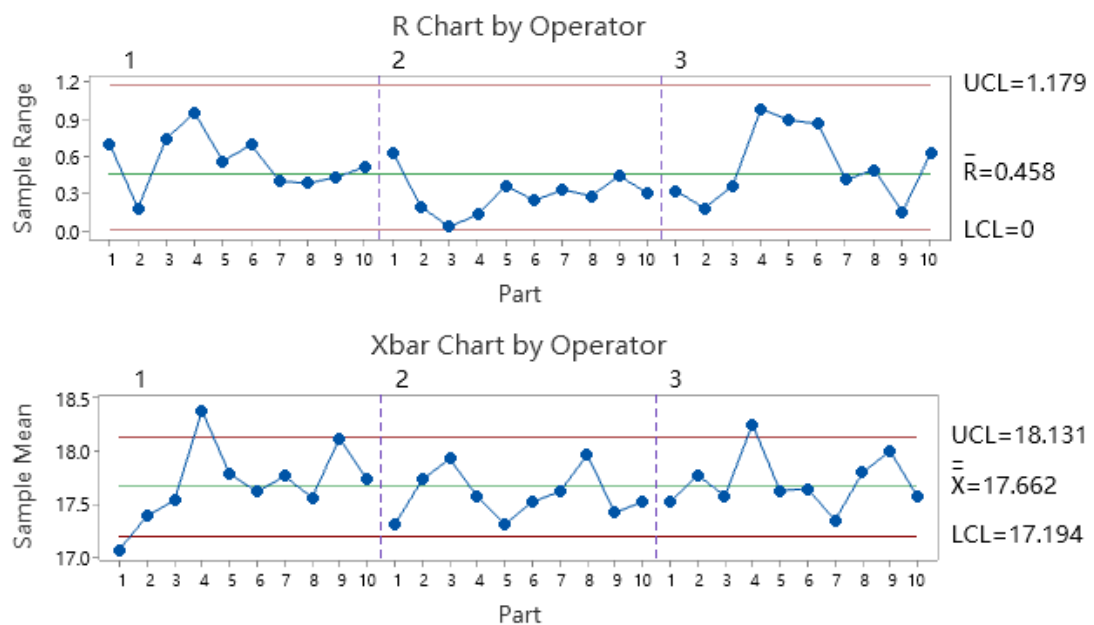


Figure10 Report for Length

Gage R&R (ANOVA) for Length

Gage name:
Date of study:

Reported by:
Tolerance:
Misc:

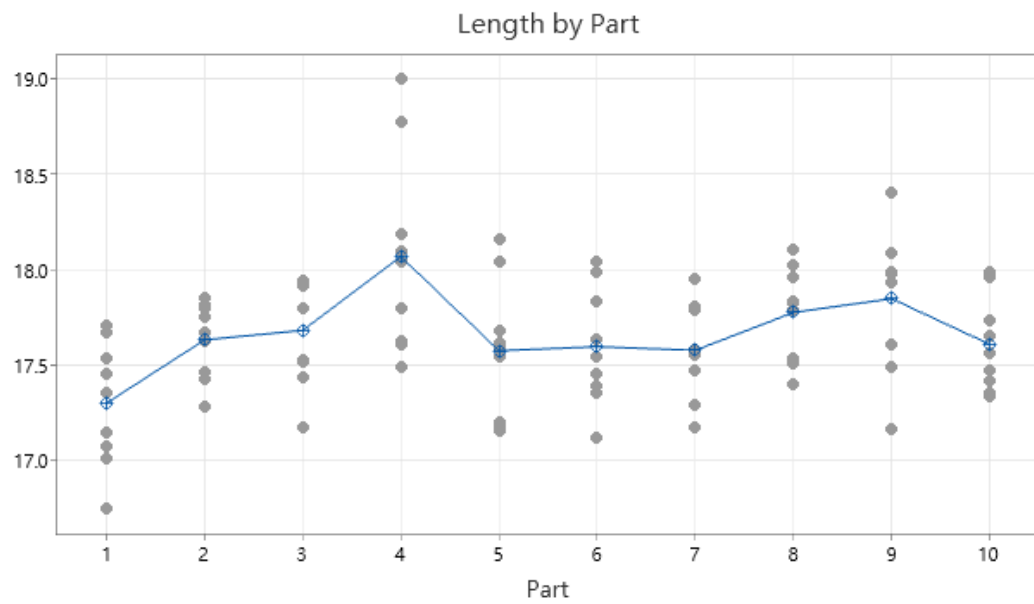


Figure10 Report for Length

Gage R&R (ANOVA) for Length

Gage name:
Date of study:

Reported by:
Tolerance:
Misc:

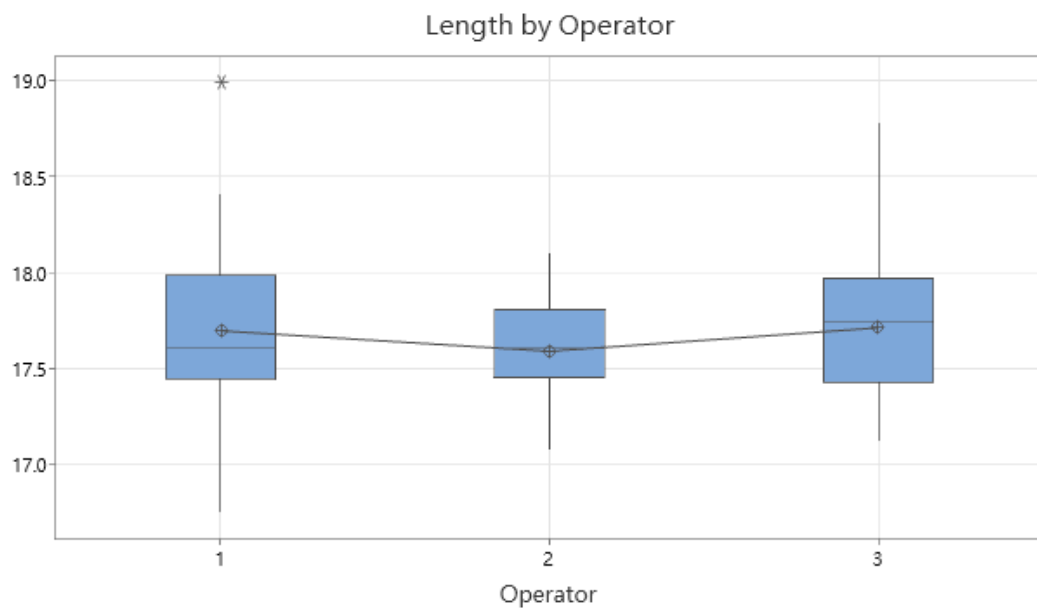


Figure10 Report for Length

Gage R&R (ANOVA) for Length

Gage name:
Date of study:

Reported by:
Tolerance:
Misc:

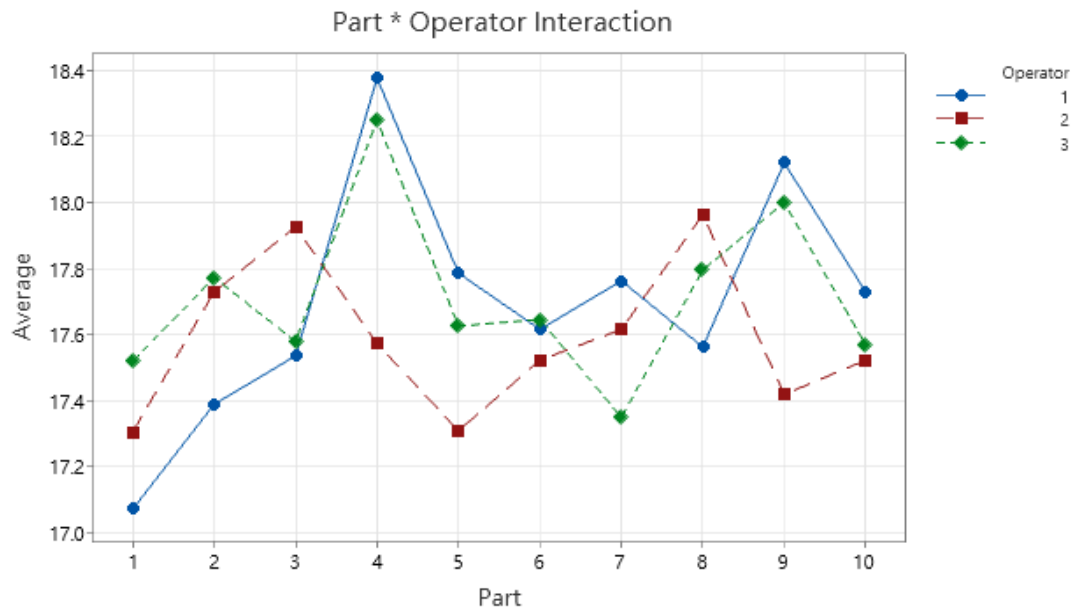


Figure10 Report for Length

Width: Unacceptable (GRR = 46.31%, ndc = 2); Part significant ($p < 0.001$), Operator significant ($p = 0.001$), Interaction not significant ($p = 0.937$); Variance composition (Contribution %) shows GRR 21.45% (EV 18.22%, AV 3.23%), PV 78.55%; R chart shows significant fluctuations, Xbar is parallel but misaligned, indicating insufficient repeatability and operator variability

Two-Way ANOVA Table With Interaction

Source	DF	SS	MS	F	P
Part	9	1.31698	0.146331	67.9750	0.000
Operator	2	0.04641	0.023203	10.7786	0.001
Part * Operator	18	0.03875	0.002153	0.5207	0.937
Repeatability	60	0.24807	0.004134		
Total	89	1.65020			

α to remove interaction term = 0.05

Figure11 Two-Way ANOVA For Width

Gage R&R

Variance Components

Source	VarComp	%Contribution (of VarComp)
Total Gage R&R	0.0043280	21.45
Repeatability	0.0036771	18.22
Reproducibility	0.0006509	3.23
Operator	0.0006509	3.23
Part-To-Part	0.0158504	78.55
Total Variation	0.0201784	100.00

Process tolerance = 2

Gage Evaluation

Source	StdDev (SD)	Study Var (6 × SD)	%Study Var (%SV)	%Tolerance (SV/Toler)
Total Gage R&R	0.065788	0.394725	46.31	19.74
Repeatability	0.060639	0.363836	42.69	18.19
Reproducibility	0.025512	0.153073	17.96	7.65
Operator	0.025512	0.153073	17.96	7.65
Part-To-Part	0.125898	0.755391	88.63	37.77
Total Variation	0.142051	0.852304	100.00	42.62

Number of Distinct Categories = 2

Figure12 Variation and Evaluation for Width

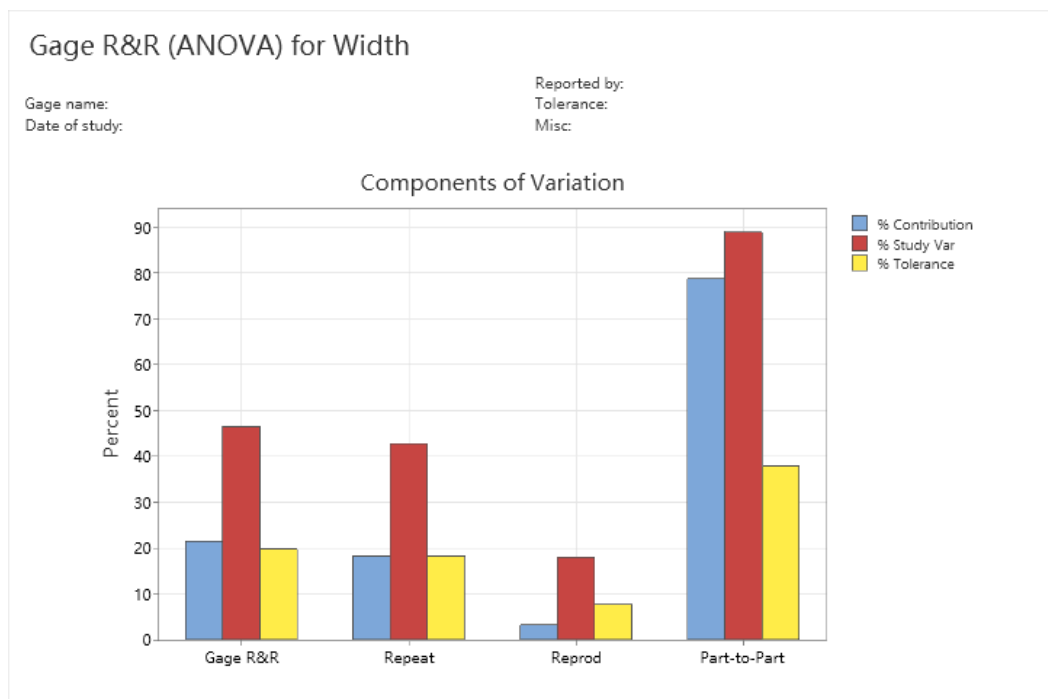


Figure13 Report for Width

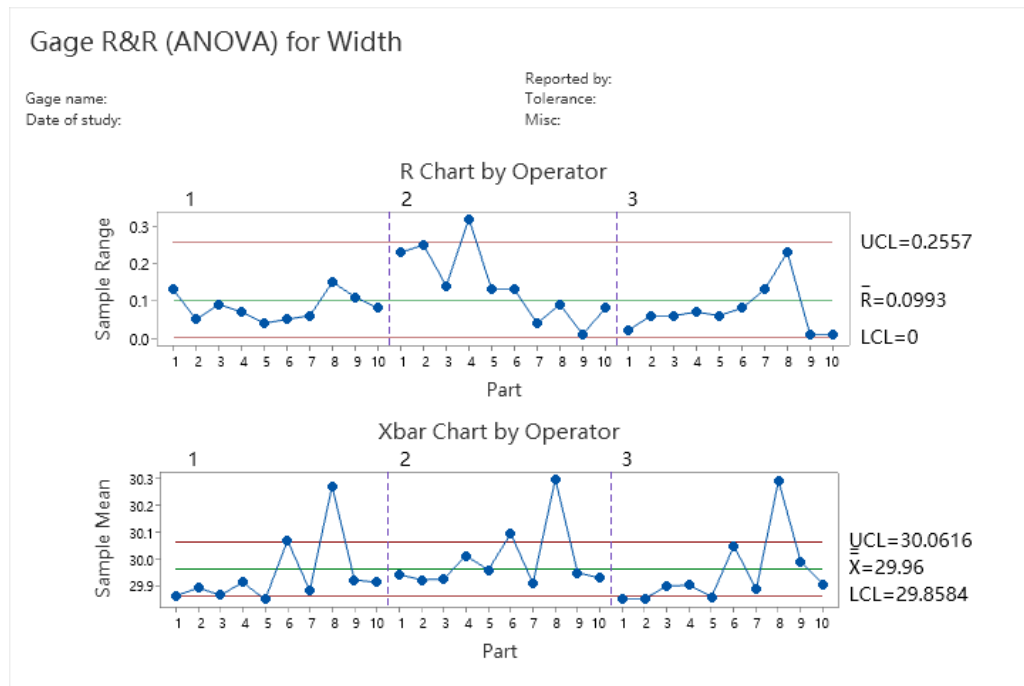


Figure13 Report for Width

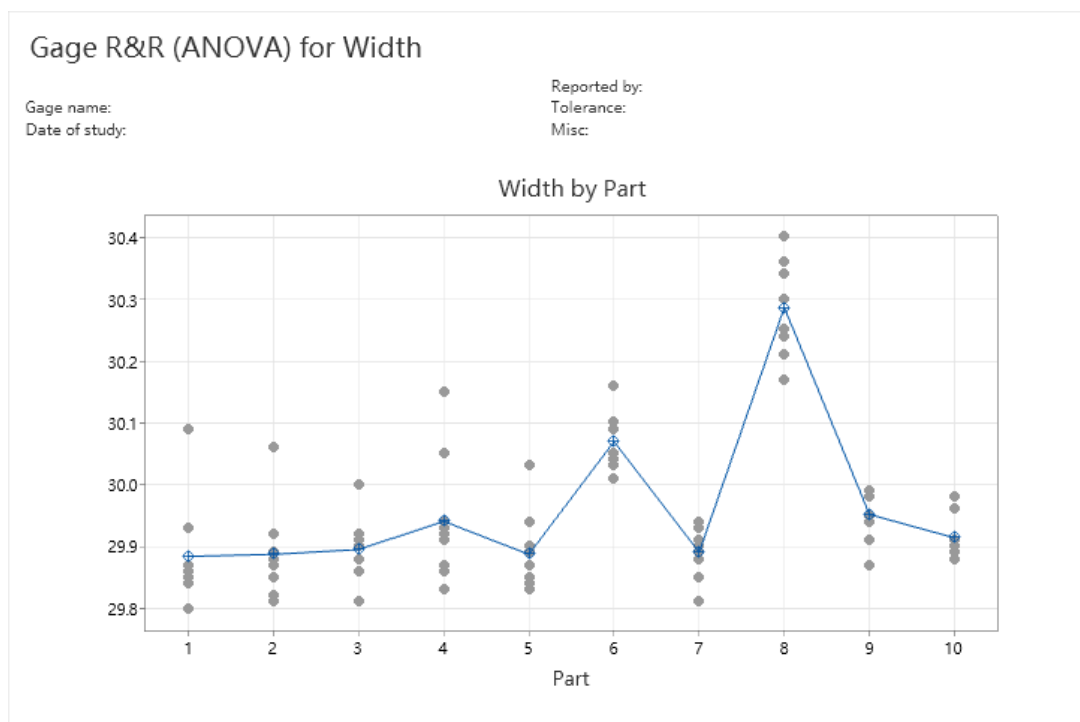


Figure13 Report for Width

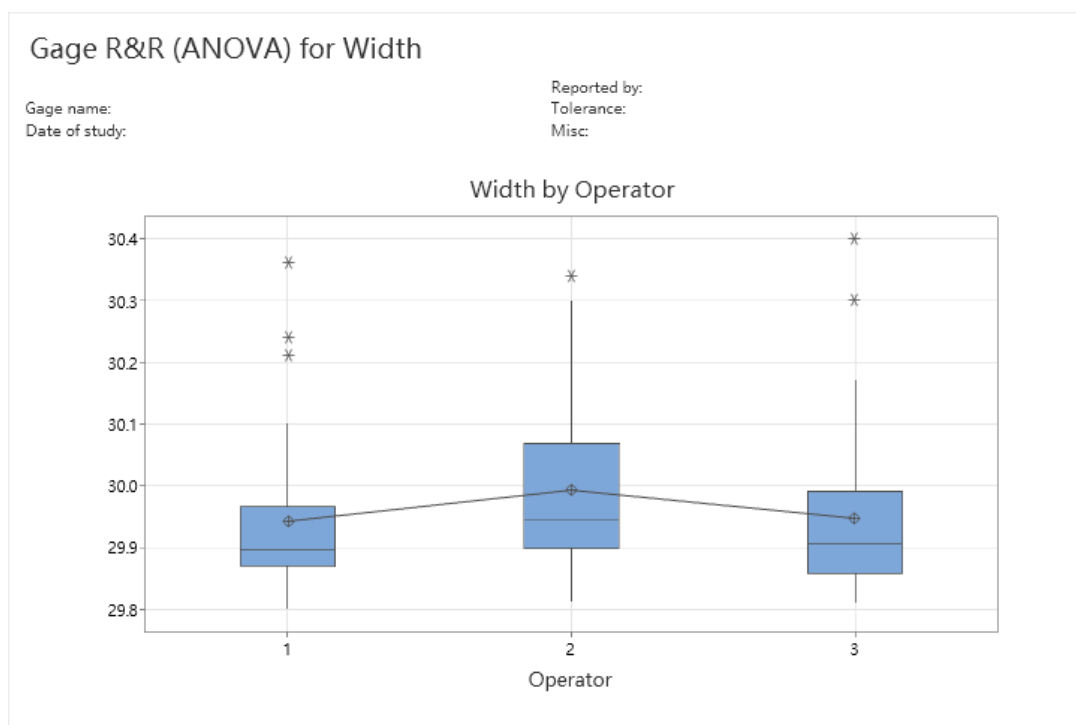


Figure13 Report for Width

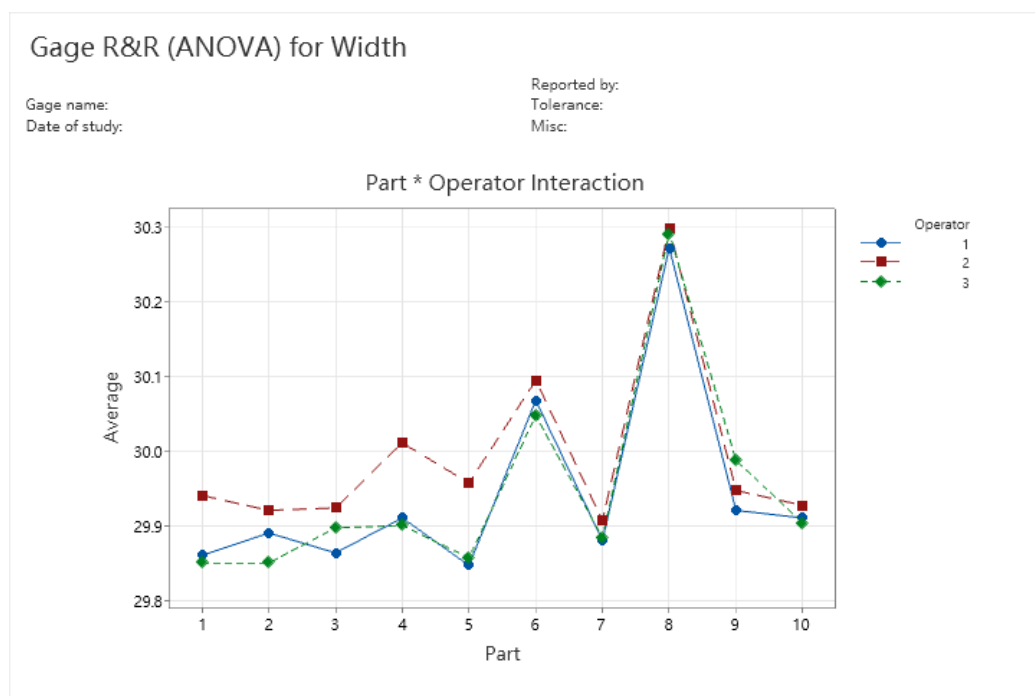


Figure13 Report for Width

Problems:

Part	Problem
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Thickness	Qualified (%GRR = 9.81%, ndc = 14; PV $\approx$ 99%, EV dominant, AV $\approx$ 0). Variations primarily stem from repeatability and resolution. Recommendations include using a micrometer with a ratchet, fixtures for fixed positioning, and ‘one person per fixture.’ Perform zero-point inspection at the start of each round and measure each point three times.
Length	Unacceptable (GRR = 92.63%, ndc = 1; significant interaction p = 0.003, EV dominant, R plot near UCL); main causes: end face fit/axial alignment and force application inconsistency, lack of end face limiting; recommend adding double end limits + V-shaped support or height gauge + platform, standardising commands and re-measuring with the same tool, and re-performing the simplified GRR.
Width	Unacceptable (GRR = 46.31%, ndc = 2; Part significant, Operator deviation p = 0.001, no interaction), R is too large and parallel to Xbar but misaligned, indicating insufficient repeatability + operator deviation; recommend adding lateral guidance/positioning, standardising light force and cross-section, and performing three measurements at the same point with the same tool. Re-test GRR after modification

Process Capability Analysis

Part	Analysis	Improve
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Thickness	Thickness (LSL = 0.70, USL = 0.90): Based on <i>Thk_Avg</i>, the capability assessment yields Ppk = 0.35 and Cpk = 0.27 (lower limit dominant), both significantly below 1.33; the mean value is approximately 0.7675 mm, skewed toward LSL, with low-side failures predominating. The process capability is deemed insufficient.	Use XY compensation/flow rate fine-tuning to adjust the average thickness to approximately 0.800 mm, then standardise layer height/outer wall/cooling to reduce fluctuations to $\sigma \leq 0.025$. Repeat the process with a minimum of ≥ 25 pieces to analyse capability, iterating until Cpk ≥ 1.33 is achieved.
Length	Capability failure — mean value 17.662 mm > USL 15.50, Ppk = – 3.58, Cpk = –4.05, 10/10 samples out of specification; additionally, the MSA for this KPI has not passed, and the dimensions are inconsistent with specifications. The conclusion is for reference only; the measuring tools must be repaired or dimensions aligned before retesting.	Use the dual-end limit + V-shaped support/height gauge method to standardise measurements, ensuring GRR $\leq 30\%$ and ndc ≥ 5, and verify the reference dimension Length = 15 ± 0.5 mm; If this is indeed the feature, modify the CAD or perform a minor XY negative compensation to bring the average back to 15.00 mm, standardise layer height/outer wall/cooling to reduce σ, and finally repeat the analysis with ≥ 25 pieces until Cpk ≥ 1.33.
Width	Wid_Avg displays the average value of 29.96 mm, Ppk $\approx$	First, ensure that the measurement system is properly calibrated.

	2.51/Cpk $\approx$ 2.62, and PPM $\approx$ 0, indicating sufficient capability; however, since the MSA for this KPI has not passed (GRR = 46.31%, ndc = 2), this conclusion is for reference only. The measurement must be corrected and verified with a sample size of ≥ 25 pieces.	Then, stabilise the printing parameters to maintain the centre at approximately 30.00 mm (and perform repeatability analysis with a sample size of ≥ 25 pieces). The target specifications are Cpk/Ppk $\geq$ 1.33 and PPM $\approx$ 0.
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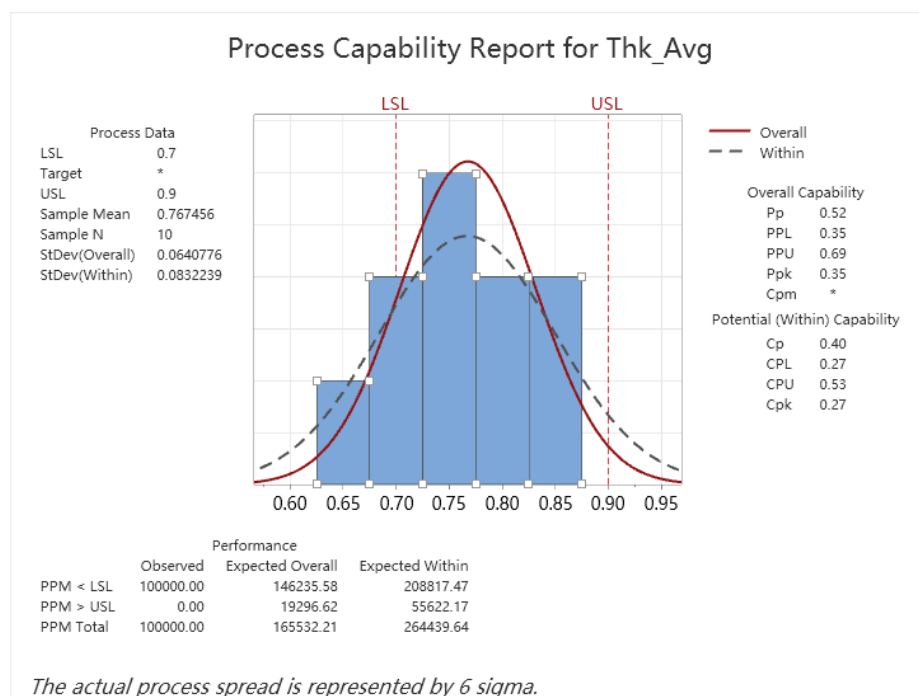


Figure14 Process Capability Report For Thickness

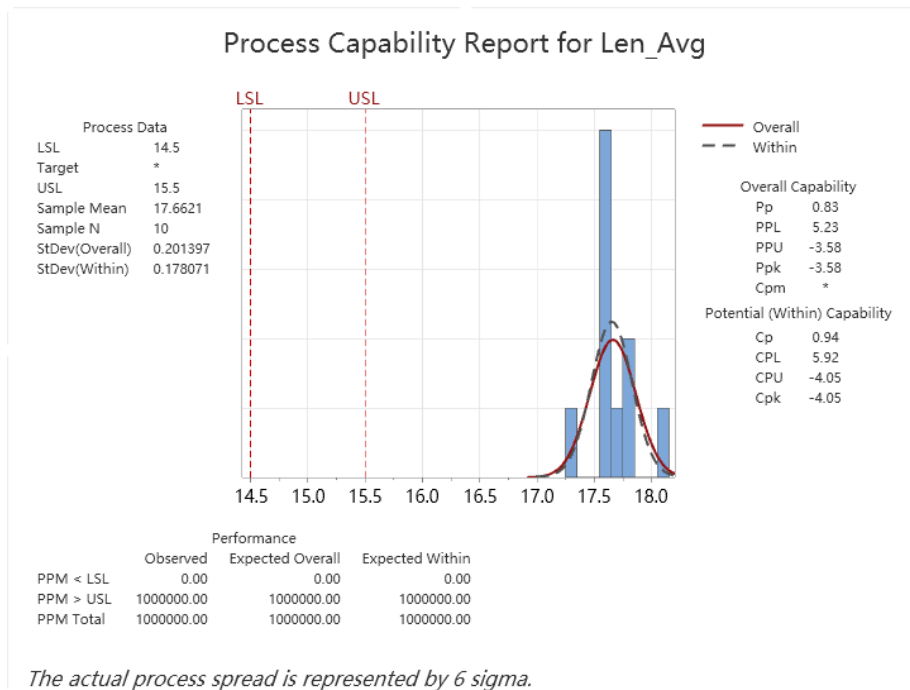


Figure 15 Process Capability Report For Lengths

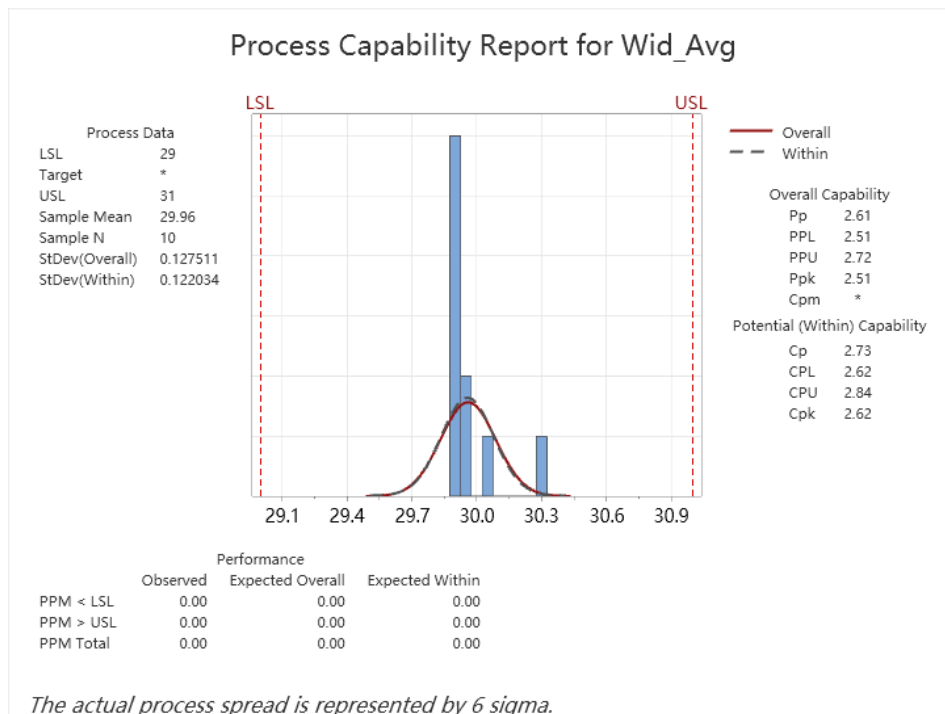


Figure 16 Process Capability Report For Width

Probability, Reliability and Quality

Part B: Quality

Name: Zhifeng Song

Summary

This report completes the measurement and capability closed-loop assessment of 3D-printed hinges: MSA indicates Thickness is usable (GRR $\approx$ 9.8%, ndc=14), Width is weak ($\approx$ 46%, ndc=2), and Length is failed ($\approx$ 92.6%, ndc=1, with significant interaction); Capability aspects: Thickness $P_{pk} \approx 0.35$ (approaching LSL), Length $P_{pk} \approx 0.56$ (constrained by USL, predicted non-conformance $\approx$ 4.7%), Width $P_{pk} \approx 1.68$ (conforming) after Johnson transformation. FMEA/fishbone diagram identifies primary causes: slicing parameters, material moisture content/batch variation and drying, temperature control/operational consistency; Secondary factors: MSA and equipment calibration drift. Recommendations prioritise parameter and WI standardisation, material drying and batch traceability, closed-loop temperature control and periodic calibration, GRR improvement, integrated with tolerances/DFM and SPC. Targets: $P_{pk} \geq 1.33$, %GRR < 10, ndc $\geq$ 10.

Summary of KPI measurement data

Thickness	Msmt 1	Msmt 2	Msmt 3	Msmt 4	Msmt 5	Msmt 6	Msmt 7	Msmt 8	Msmt 9
1	0.798	0.79	0.796	0.799	0.8	0.798	0.805	0.785	0.799
2	0.739	0.742	0.732	0.737	0.738	0.721	0.725	0.735	0.736
3	0.856	0.855	0.851	0.859	0.855	0.848	0.856	0.861	0.853
4	0.758	0.759	0.751	0.758	0.752	0.748	0.754	0.758	0.743
5	0.803	0.8	0.804	0.817	0.811	0.792	0.81	0.801	0.803
6	0.72	0.724	0.732	0.731	0.717	0.713	0.733	0.733	0.717
7	0.876	0.868	0.876	0.879	0.869	0.869	0.872	0.872	0.878
8	0.673	0.665	0.666	0.679	0.674	0.665	0.671	0.685	0.682
9	0.715	0.708	0.699	0.711	0.697	0.697	0.712	0.711	0.713
10	0.754	0.744	0.758	0.762	0.753	0.748	0.75	0.752	0.757
Length	Msmt 1	Msmt 2	Msmt 3	Msmt 4	Msmt 5	Msmt 6	Msmt 7	Msmt 8	Msmt 9
1	17.01	16.75	17.45	17.07	17.14	17.7	17.35	17.53	17.67
2	17.28	17.46	17.42	17.81	17.62	17.75	17.67	17.79	17.85
3	17.91	17.17	17.52	17.94	17.91	17.92	17.43	17.51	17.79
4	18.04	18.09	18.99	17.49	17.62	17.6	18.18	17.79	18.77
5	17.59	17.61	18.15	17.54	17.2	17.18	17.68	17.15	18.04
6	18.04	17.35	17.45	17.63	17.54	17.39	17.12	17.83	17.98

7	17.78	17.95	17.55	17.57	17.8	17.47	17.29	17.58	17.17
8	17.78	17.5	17.4	17.82	18.1	17.96	17.53	17.83	18.02
9	17.98	18.4	17.97	17.6	17.49	17.16	17.98	17.93	18.08
10	17.47	17.98	17.73	17.35	17.56	17.65	17.41	17.33	17.96
Width	Msmt 1	Msmt 2	Msmt 3	Msmt 4	Msmt 5	Msmt 6	Msmt 7	Msmt 8	Msmt 9
1	29.93	29.8	29.85	30.09	29.87	29.86	29.86	29.85	29.84
2	29.88	29.92	29.87	30.06	29.89	29.81	29.85	29.82	29.88
3	29.9	29.81	29.88	29.86	29.91	30	29.91	29.86	29.92
4	29.92	29.87	29.94	30.15	29.83	30.05	29.93	29.86	29.91
5	29.84	29.83	29.87	30.03	29.9	29.94	29.85	29.83	29.89
6	30.05	30.05	30.1	30.16	30.03	30.09	30.01	30.04	30.09
7	29.91	29.85	29.88	29.93	29.89	29.9	29.94	29.81	29.9
8	30.36	30.24	30.21	30.3	30.25	30.34	30.4	30.3	30.17
9	29.91	29.87	29.98	29.94	29.95	29.95	29.99	29.98	29.99
10	29.88	29.96	29.89	29.9	29.98	29.9	29.91	29.9	29.9

Introduction



Figure 1: 10 Hinges printing

This report analyses the normality and process capability (P_p/P_{pk} , C_p/C_{pk}) of critical dimensions for ten 3D-printed hinge prototypes, quantifying current manufacturing capability and sources of variation. Based on this analysis, process flow diagrams, FMEA, and fishbone diagrams were developed. A prioritised list of design and process improvements, ranked by impact, was proposed to establish an actionable quality enhancement roadmap.

Q1. KPI Definition and Capability Assessment

Thickness:

The normality of the 10 mean thickness data points was confirmed ($AD=0.232$, $p=0.730 > 0.05$), with a mean of 0.7675 mm and $SD=0.0641$ mm. Therefore, normal distribution analysis may be conducted directly without data transformation.

Project	Result
Sample size N	10
Mean value (mm)	0.767

Standard Deviation (Overall)	0.064
Minimum/Maximum (mm)	0.673 / 0.873
Normality test	AD=0.232, p=0.730 (passed)
Specification boundaries	LSL=0.70, USL=0.90
Overall capability	$P_p=0.52$, $P_{pk}=0.35$ (PPL=0.35, PPU=0.69)
Short-term/potential capability	$C_p=0.40$, $C_{pk}=0.27$ (CPL=0.27, CPU=0.53)
Projected Non-Conformance Rate (Overall)	<LSL: $\approx 146,236$ ppm; >USL: $\approx 19,297$ ppm; Total: $\approx 165,533$ ppm
Sample observation failed to meet standards	1/10 below LSL ($\approx 100,000$ ppm)
Gauge Status (GRR)	GRR $\approx 9.8\%$, ndc=14 (available)
Conclusion	Capability is insufficient and skewed towards the lower end (close to LSL), requiring "centring and reduction of deviation".

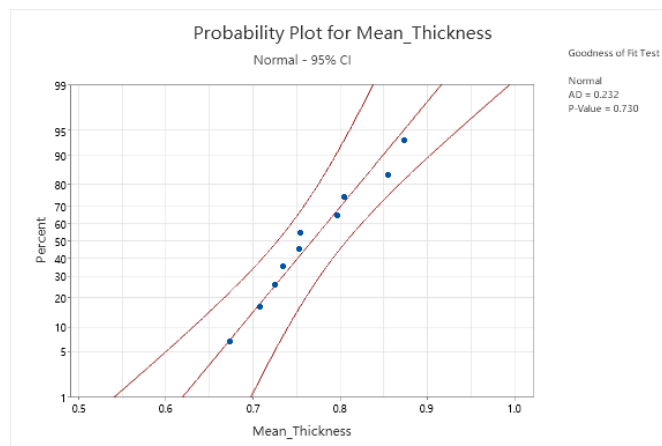


Figure 2: Probability plot for Thickness

Descriptive Statistics

N	N*	Mean	StDev	Median	Minimum	Maximum	Skewness	Kurtosis
10	0	0.767456	0.0640776	0.753278	0.673333	0.873222	0.395749	-0.677962

Goodness of Fit Test

Distribution	AD	P
Normal	0.232	0.730

ML Estimates of Distribution Parameters

Distribution	Location	Shape	Scale	Threshold
Normal*	0.76746		0.06408	

* Scale: Adjusted ML estimate

Figure 3: Distribution Identification for Thickness

Analysis Conclusions:

The thickness data exhibit near-normal distribution (AD $p=0.73$), yet process capability is markedly insufficient: $P_p \approx 0.52$, $P_{pk} \approx 0.35$ ($C_p \approx 0.40$, $C_{pk} \approx 0.27$) well below 1.33. The mean of 0.767 approaches $LSL=0.70$, with most out-of-specification variation occurring below the lower limit (observed scrap rate $\approx 10\%$). The conclusion is that the Thickness KPI currently lacks manufacturing capability.

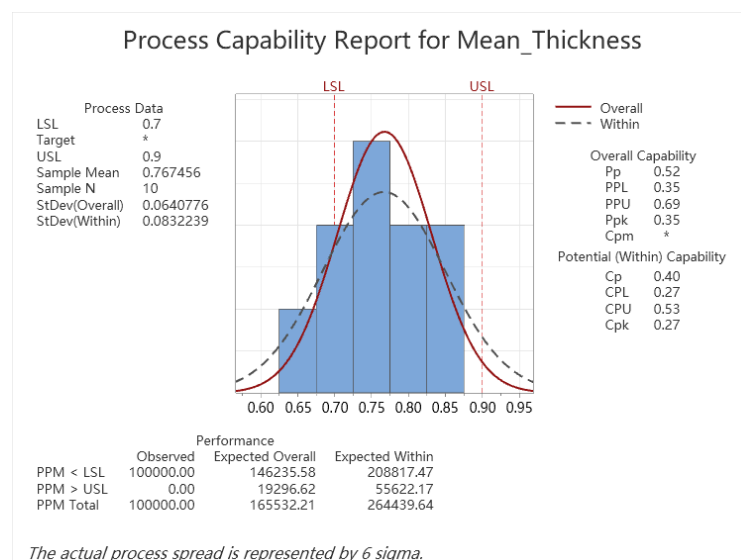


Figure 4: Process Capability Report for Thickness

Width:

Sample mean 29.96 mm (N=10), population standard deviation 0.1275 mm, range 0.402 mm

(29.883–30.286); distribution skewed to the right (Skew=2.27, Kurt=5.17).

Project	Result
Sample size N	10
Mean value (mm)	29.96
Standard Deviation (Overall)	0.1275
Minimum/Maximum (mm)	29.883 / 30.286
Normality test	AD = 1.439, $p < 0.005$ (failed, non-normal)
Specification boundaries	LSL = 29.00, USL = 31.00
Overall capability	$P_p=2.98$, $P_{pk}=1.68$ (PPL = 44.63, PPU = 1.68)
Short-term/potential capability	Johnson transformation reports only Overall; normal fit for reference: $C_p=2.73$, $C_{pk}=2.62$ (CPL=2.62, CPU=2.84)
Projected Non-Conformance Rate (Overall)	<LSL: 0 ppm; >USL: 0 ppm; Total $\approx$ 0 ppm
Sample observation failed to meet standards	0/10 Exceeds specifications
Gauge Status (GRR)	%GRR $\approx$ 46%, ndc = 2 (repeatability somewhat weak, operator variation present)
Conclusion	The overall dimensional capability is favourable and well-removed from specification limits ($P_{pK} \approx 1.68$), though the distribution is non-normal and the measurement system analysis (MSA) is weak. Prior improvement in measurement consistency is required, upon which process stability can be maintained.

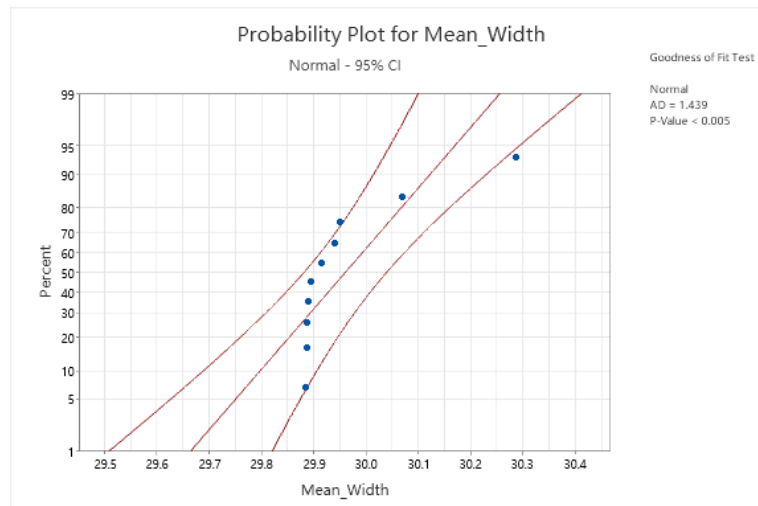


Figure 5: Probability plot for Width

Descriptive Statistics

N	N*	Mean	StDev	Median	Minimum	Maximum	Skewness	Kurtosis
10	0	29.96	0.127511	29.9039	29.8833	30.2856	2.26577	5.16637

Goodness of Fit Test

Distribution	AD	P
Normal	1.439	<0.005

ML Estimates of Distribution Parameters

Distribution	Location	Shape	Scale	Threshold
Normal*	29.96000		0.12751	

* Scale: Adjusted ML estimate

Figure 6: Distribution Identification for Width

Analysis Conclusions:

Width data did not follow a normal distribution (AD $p < 0.005$, skewness ≈ 2.27 , kurtosis ≈ 5.17). Following Johnson transformation assessment, process capability was deemed adequate: $P_p \approx 2.98$, $P_{pk} \approx 1.68$ (> 1.33), with $PPL \approx 44.6$ and $PPU \approx 1.68$ indicating the upper side (near USL) as the limiting side. The mean of 29.96 mm lies midway within the 29–31 mm specification, with a predicted non-conformance rate of ≈ 0 ppm. Conclusion: The Width KPI now exhibits good capability. Should further improvement be desired, consider reducing upper-side variation or slightly shifting the mean downward to elevate P_{pk} .

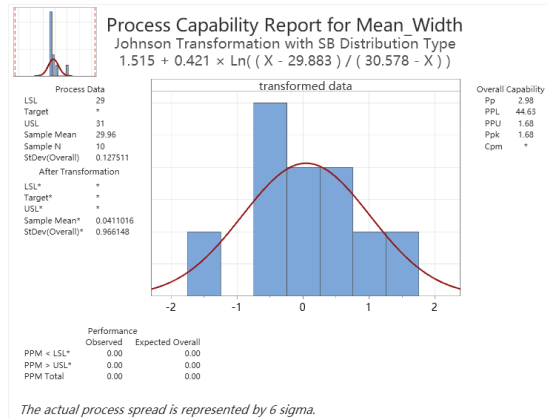


Figure 7: Process Capability Report for Width

Length:

Length data approximates a normal distribution (AD $p=0.221$), yet process capability is inadequate and skewed towards the upper limit: $P_p \approx 0.83$, $P_{pk} \approx 0.56$ (approaching $USL=18$). One-tenth of the sample exceeds the upper limit, with the overall model projecting a non-conforming rate of approximately 47k ppm.

Project	Result
Sample size N	10
Mean value (mm)	17.662
Standard Deviation (Overall)	0.201
Minimum/Maximum (mm)	17.297 / 18.063
Normality test	AD = 0.447, p = 0.221 (passed)
Specification boundaries	LSL = 17.00, Target = 17.50, USL = 18.00
Overall capability	$P_p=0.83$, $P_{pk}=0.56$ (PPL = 1.1, PPU = 0.56)
Short-term/potential capability	(Within/ref.) $C_p \approx 0.83$, $C_{pk} \approx 0.56$ (not explicitly reported by output)
Projected Non-Conformance Rate (Overall)	<LSL: ≈ 505 ppm; >USL: $\approx 46,701$ ppm; Total: $\approx 47,206$ ppm
Sample observation failed to meet standards	1/10 above USL
Gauge Status (GRR)	%GRR $\approx 92.6\%$, ndc = 1; operator $\times$ part

	interaction significant → MSA unacceptable
Conclusion	This process exhibits defects and exhibits a significant bias towards the upper specification limit (USL). First, correct the MSA (repeatability/interaction), then reposition to the target and reduce variability.

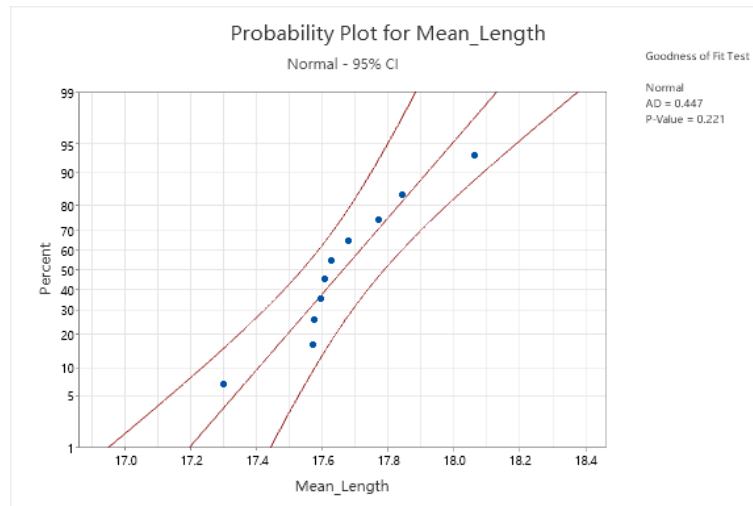


Figure 8: Probability plot for Length

Descriptive Statistics

N	N*	Mean	StDev	Median	Minimum	Maximum	Skewness	Kurtosis
10	0	17.6621	0.201397	17.6161	17.2967	18.0633	0.357168	1.64864

Goodness of Fit Test

Distribution	AD	P
Normal	0.447	0.221

ML Estimates of Distribution Parameters

Distribution	Location	Shape	Scale	Threshold
Normal*	17.66211		0.20140	

* Scale: Adjusted ML estimate

Figure 9: Distribution Identification for Length

Analysis Conclusions:

Approximately normal ($p=0.221>0.05$). At LSL=17, target 17.5, and USL=18, the mean of 17.662 is above the upper limit; $P_p \approx 0.83$, $P_{pk} \approx 0.56$ (constrained by upper limit), with an estimated non-conformance rate of $\approx 4.7\%$ ($\approx 47k$ ppm). Conclusion: Insufficient capability

with tendency towards upper limit. Recommendations: 1. Centre mean value to ≈ 17.5 ; 2. Reduce variability (constant speed/extrusion/temperature control, optimise layer height and path); 3. If GRR is excessive, prioritise MSA improvement.

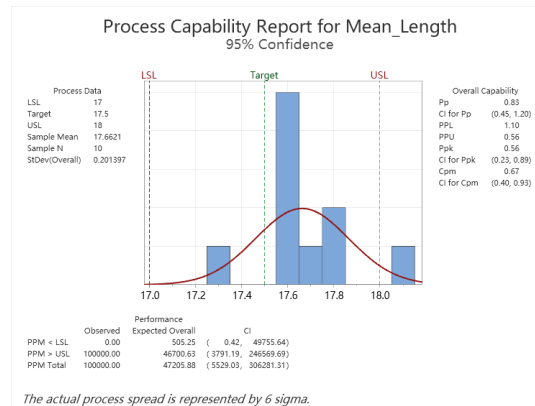
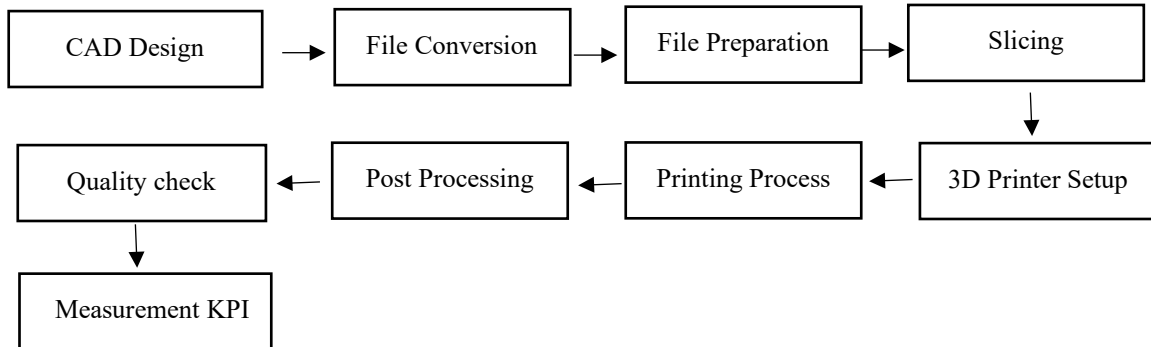


Figure 10: Process Capability Report for Length

Q2 Capability Analysis and Root Cause Identification in 3D Printing Processes

2.1 3D Printer Process map:



The process iterates through slicing and setting phases, adjusting parameters to ensure quality. Subsequently, its effectiveness is evaluated through KPI measurement.

2.2 FMEA

Failure mode	Potential impact	Potential causes/mechanisms	Current control	S	O	D	RPN
Incorrect slicing parameter settings (layer height/wall thickness/scaling factor, etc.)	Thickness deviation from target, poor surface quality, increased dimensional dispersion	Lack of experience in parameter selection; standard templates were not reused.	Use default template, first article inspection	8	6	4	192
Platform not levelled/Nozzle Z-axis offset incorrect	Poor adhesion or flattening at the base layer, resulting in thickness/length deviations.	Failure to recalibrate after material change/maintenance	First-floor test panels, routine levelling	8	5	3	120
Temperature control fluctuations	Shrinkage and warping,	PID not tuned, fan/ambient	Enclosed chassis, PID	7	5	5	175

(nozzle/heated bed/environment)	length/thickness instability	temperature fluctuations	self-tuning, temperature monitoring				
Step/Extrusion Calibration Deviation (XY/Z/Extrusion Volume)	Systematic dimensional deviation (e.g. Length near USL, Thickness near LSL)	Stepper/extruder uncalibrated or drifting	Monthly calibration, calibration records	9	4	4	144
Material moisture content/batch variation	Bubbles, surface defects, and shrinkage lead to increased dimensional variation.	Improper storage of consumables/Insufficient drying	Drying process, batch traceability	6	6	5	180
Operational inconsistencies/Unclear work instructions	Significant variation within the same cohort, with declining capabilities Significant variation within the same cohort and reduced capabilities	Inadequate training/Incomplete handover information	Work Instructions, Shift Handover Checklist	6	5	6	180
Measurement system instability (fixtures/zero position/contact force)	Misjudgements or biased rulings, resulting in	Measuring tools are uncalibrated and positioning is non-repeatable.	Periodic calibration of measuring	7	4	6	168

	distorted assessments of capability		instruments, GRR assessment				
Post-processing (deburring/polishing) Intrusion dimension	Thickness/length altered, measurement deviation	Unlimited position or excessive operation	Limit fixture, post- processing checklist	5	4	6	120

Table1: FMEA (This approach employs $RPN \geq 150$ or $S \geq 8$ as the priority improvement threshold (referencing common industry practices and accounting for sample size).

Focus on the most critical issues (such as improper slicing parameter settings), prioritising the elimination of root causes with the greatest impact on product quality. This approach enhances process stability and reliability while improving the consistency of finished products.

2.3 Fishbone

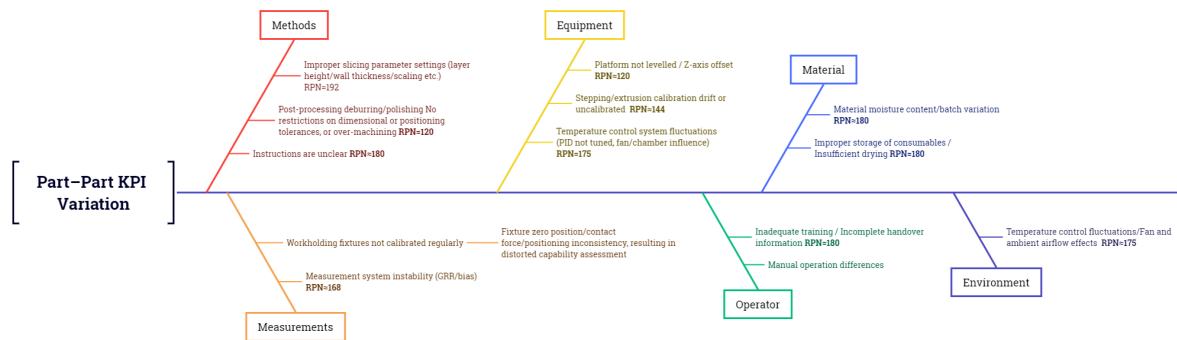


Figure 11: Fishbone

Using a combined FMEA and fishbone diagram assessment, the current dimensional variation is primarily influenced by slicing parameters, material moisture content/batch, and operational and temperature control factors, with measurement and equipment calibration being secondary contributors. It is recommended to prioritise the following improvements based on $RPN \geq 150$ or $S \geq 8$: parameter standardisation, material drying traceability, closed-loop temperature control, and GRR enhancement. This will systematically enhance process capability and consistency.

Prioritized list

Priority	Type	Change	Why (Associated	Verify (measurement/method)
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			risks/mechanisms)	
1	Process	Standardised slicing parameters: layer thickness, wall thickness, infill rate/pattern, print speed, and top/shell infill. Establish machine model × material templates with version control.	Inappropriate selection of current parameters → dimensional variation and surface defects; Relying solely on default templates/first-piece inspection proves insufficient for prevention.	Comparison of C_p/C_{pk} , P_{pk} and defective ppm before and after AB testing; Significance of analysis of variance/regression tests
2	Process	Material Drying and Batch Traceability: Feed drying curve, moisture content inspection, material roll sealing and batch-printing records.	Moisture content/batch variations causing bubbles, shrinkage and dimensional shifts	Record moisture content (%) and correlate/regress against dimensional deviation; Compare C_p/C_{pk} within and across batches.
3	Process	Temperature control closed loop: Nozzle/heated bed PID tuning, enclosed chassis + directional air ducts; Set "temperature stability criteria" before commencing printing.	Temperature/airflow fluctuations causing interlayer shrinkage differential and warping	Collect temperature time series σ and regress against dimensional deviation; P_{pk} improvement ≥ 0.2 shall be deemed compliant.
4	Process	Equipment periodic calibration: Stepping/extrusion volume, Z-offset/platform levelling (monthly); Electronic calibration records	Calibration drift and platform unevenness cause systematic bias.	The mean difference and standard deviation decreased between the pre-calibration and post-calibration batches; the X-bar/R or I-MR chart remained stable.
5	Process	Work Instruction Consistency and	Human operational variations and unclear communication	WI audit pass rate: Reduction in shift-to-shift mean difference;

		WI/Training: Key Steps Work Instructions, Change Approval, First Article Verification; Team Rotational Training and On-the-Job Verification	lead to fluctuations in capability.	Decrease in rework/scrap rate
6	Process	Measurement System Analysis (MSA)/Gage Repeatability and Reproducibility (GRR) Improvement: Gage calibration, fixture positioning, standardised contact force/reference; Replace digital readout gauges where necessary.	MSA instability may obscure genuine improvements and amplify judgemental fluctuations.	%GRR <10%, ndc ≥10; repeatability/reproducibility meets standards
7	Design	Tolerance and Reference Optimisation: Specifying manufacturable tolerances for KPI features; standardising positioning/measurement references	Excessively tight or ambiguous tolerances lead to "apparent fluctuations" and disputes.	Tolerance and Process Capability Matching: $C_p/C_{pk} \geq 1.33$; Inspection Fixture Consistency Validation
8	Design	Design for Manufacturability (DFM): Add chamfers/fillet where necessary, incorporate localised thickening or ribs, and optimise overhang/bridge angles.	Reduce shrinkage/warping and minimise post-processing intervention risks	Design AB trials using identical machinery and materials before and after modification; statistical comparison of flatness/dimensional deviations

9	Process	Optimisation of Placement and Path Strategies: Component Orientation/Layout, Support Strategy, Minimisation of Thermal Interference Between Co-located Components	Temperature non-uniformity and path-induced dimensional drift	Comparison of temperature fields and dimensional variations across solutions, with the highest P_{pk} value achieved in the curing process
10	Process	First article – process – finished part three-point quality inspection: First article release upon acceptance, process sampling to monitor drift, finished part critical dimensions full inspection.	End-point detection alone is costly and prone to false negatives.	Establish SPC control charts and deviation rules; Offset → Re-inspection/Correction; ppm reduction

Prioritise speed over stability: Focus first on advancing items 1–4 and 6 (slice parameters, drying/traceability, closed-loop temperature control, equipment calibration, MSA improvement) to rapidly elevate P_{pk} / C_{pk} ; concurrently implement items 7–8 (tolerance and DFM optimisation), embedding improvements into WI/training. Throughout, implement PDCA/DMAIC with small-batch AB validation and data closed-loop. Weekly kanban monitoring of C_p / C_{pk} , P_{pk} , ppm, %GRR, temperature σ , and moisture content. Trigger corrective action and revalidation if $P_{pk} < 1.33$ or %GRR > 10%.

Probability, Reliability and Quality

Reliability

Name: Zhifeng Song

Summary

Overall assessment: The measurement system is compliant, with $n=10$ for each sample. The lifespan disparity between New and Old manifests primarily in ‘early phase versus long tail’: Weibull fitting reveals New $\beta=4.06>1$ (wear-dominated, increasing risk over time), Old $\beta=0.55<1$ (early defect-dominated, more stable survivors with a long tail); Thus B10 significantly outperforms Old (17.9 cycles vs 1.08 cycles, New exhibits higher early reliability), yet lags markedly in MTTF and median lifetime (28.2 cycles vs 110.3 cycles, Old ≈ 3.9 times more durable). Non-parametric tests confirm tail dominance: Log-rank ≈ 0.051 , Wilcoxon ≈ 0.438 also supports ‘tail-end disparity as primary factor’. Size-lifespan correlation is insignificant; core issues lie in process: first-layer adhesion/seams, elephant feet/XY compensation, cooling and material moisture content. Recommendations include Flow/EM adjustment upwards, first-layer and seam control, outer wall $\times 4$ + layer height 0.12, progressive temperature/air cooling, material drying and traceability. These measures should enhance late-stage durability while maintaining New's low early failure rate. Due to limited sample size, further validation with 20–30 additional samples is advised to narrow confidence intervals.

Design & Process Changes for 3D-Printed Hinges

Measurement data

Old Hinges:

Part	Thickness	Width	Length	Cycles
1	0.79666667	29.8833333	17.2966667	73
2	0.73388889	29.8866667	17.6277778	252
3	0.85488889	29.8944444	17.6777778	1
4	0.75344444	29.91375	18.135	338
5	0.80455556	29.86875	17.5711111	75
6	0.72444444	30.0688889	17.5922222	104
7	0.87322222	29.89	17.5733333	3
8	0.67333333	30.2855556	17.7711111	1
9	0.707	29.9511111	17.87375	1
10	0.75311111	29.9133333	17.6044444	130

New Hinges:

Part	Thickness	Width	Length	No of Bends Till Failure
1	0.708	29.95	16.39	28
2	0.701	29.94	16.5	15
3	0.723	29.89	16.33	37
4	0.691	29.89	16.1	27
5	0.708	29.9	16.36	23
6	0.698	29.98	16.38	23
7	0.719	29.9	16.27	42
8	0.694	29.94	16.27	36
9	0.706	29.94	16.32	27
10	0.713	29.91	16.24	24

List of improvements

Priority	Improvements	Factual Evidence	Objectives and Parameters	Mechanisms and expected impacts	Validation metrics
P1	Thickness centred (Flow/EM calibration)	New Thickness 0.691-0.723, mean ≈ 0.707 , below Target 0.72, SD ≈ 0.010 (very steady but low)	Flow/EM +1 to 3%; target actual thickness 0.73-0.74 mm; keep first layer extrusion stable	Brace the mean from the LSL side to near the target, $C_{pk} \uparrow$, early failure risk $\downarrow$	$C_{pk}(\text{Thickness}) \geq 1.33$; mean ≈ 0.73 -0.74, SD maintained ~ 0.01
P2	Length Back to Specification Tape (XY Dimensional Compensation/Step Calibration/Elephant's Foot Elimination)	New Length 16.10-16.50 in many places> USL=16.00 (Old larger 17-18)	Enable Elephant Foot Compensation 0.15-0.25 mm; XY Size Compensation - 0.2%~0.8% (try in small steps first); calibrate X/Y steps/mm if necessary; if the model is nominally large, CAD back to change.	First-order solution to the systematic deviation caused by the first-floor flare with a slightly larger XY, pulling the length back to ≤ 16	Mean length ≤ 15.8 -16.0; number of excesses = 0; $C_{pk}(\text{Length}) \geq 1.33$
P3	Wall/crusting	The fatigue lifespan of this cohort is centred on 15-42 episodes, with the longevity	Perimeters = 4 (promoted if currently <4; retained if already 4)	Wall number $\uparrow \rightarrow$ interlayer shear load $\uparrow$, insensitive to flow/thickness microwave motion,	$t_{10} \geq 30$ for the next round of Cycles; extreme early deaths = 0

		tail disappearing		high lifetime tail return	
P4	Step down from the first floor	0.15 mm QUALITY Placement	Maintain 0.15 mm; for a further step up in life expectancy, 0.12 mm	Layer height ↓ → better interlayer integration, defect size ↓, more stable fatigue life distribution	Capacity is maintained (C_p does not decrease), the distribution of Cycles is shifted to the right.
P5	XY Orientation (confirmation)	Preview shows that it has been placed horizontally (should be hinge axis in XY)	Hold XY; fix Seam to non- functional side	Avoids Z-direction delamination tearing; controlled surface seams	Failure mode changed from "interlayer peeling" to "high frequency fatigue" characteristics; uniform appearance of seams
P6	Cooling strategy (small cross section)	No clear fan strategy seen in preview	Small cross- section Fan 50- 70%; bridging retains high winds, non-bridging moderate	Overcooling causes cold joints and brittle layers; fan curve optimisation reduces microcracking	Appearance streaking/weakening of cold joints; Thickness C_p maintained or ↑
P7	Drying & Storage	Unrecorded batches/water content	55-60°C x 4-6h before printing; dry box storage	Water content → poor stomatal/interlaminar bonding → early death; drying can converge dispersion	$\sigma \downarrow$ for Cycles; early failure samples = 0
P8	Measurement and	Last assessed as	Calliper	Traceability to avoid	Measurement

	Processes (Gauges/Marking)	"incomplete chain of evidence/mixed sample"	calibration; each H1-H10 marked	mixing of samples; measurement repeatability ↑	review difference ≤0.02 mm; complete data sheet
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Slicing parameters

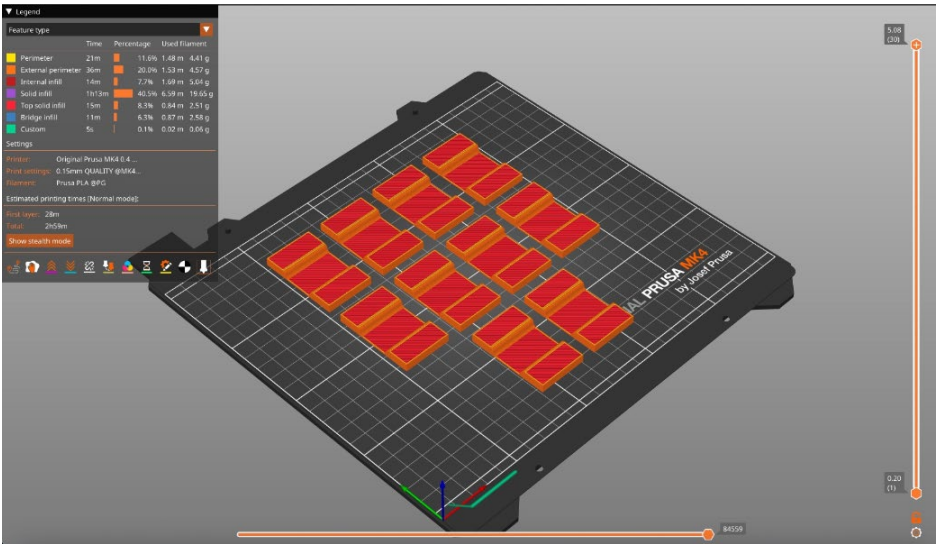


Figure 1: Platform Layout and Time Consumption Preview

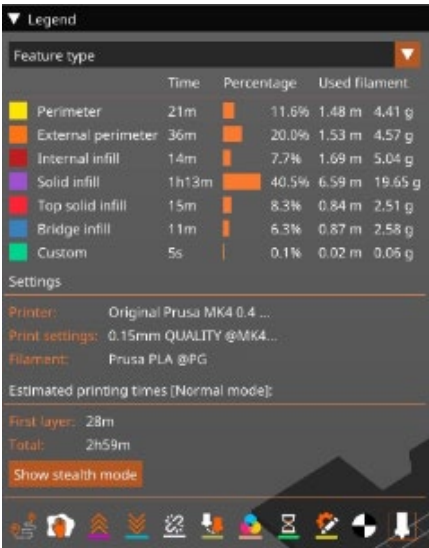


Figure 2: Slicing Time and Consumables Breakdown

Printing and Numbering

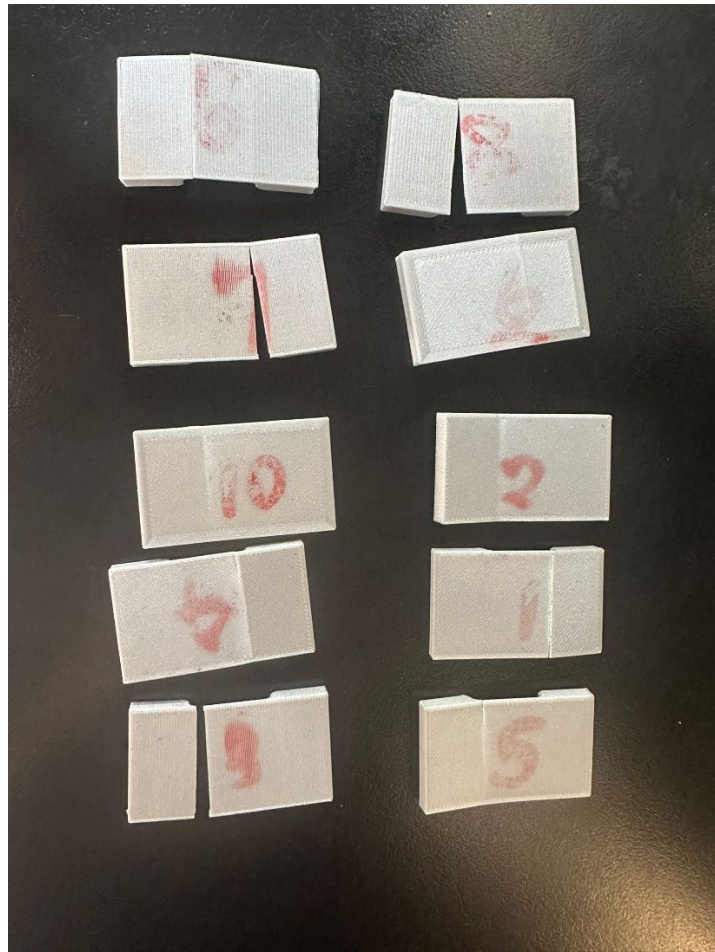


Figure 3: Old hinges after bending tests

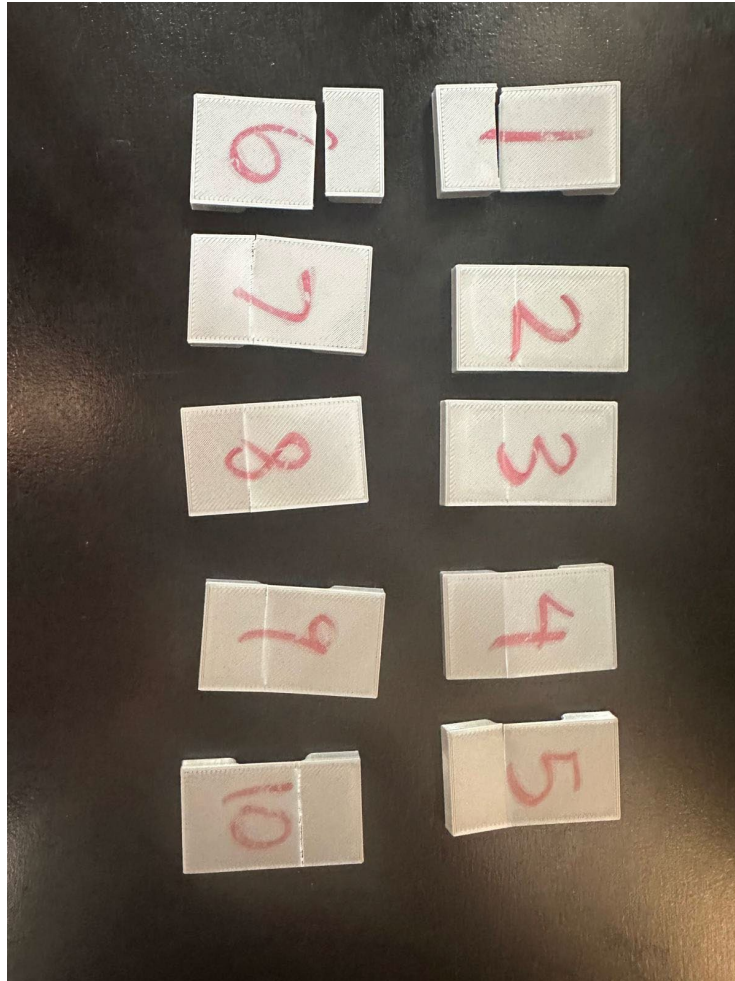


Figure 4: New hinges after bending tests

KPI Inspection, Competency Analysis, Part Quality

Histogram + Fitting

Thickness:

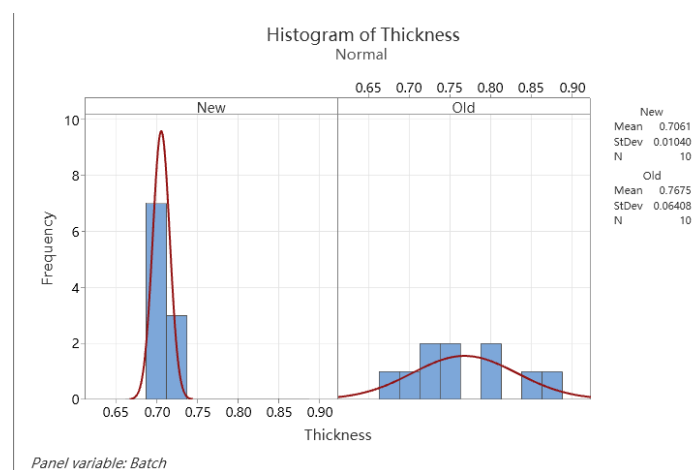


Figure 5: Histogram of Thickness

Thickness: new batch average 0.706 mm, $\sigma=0.010$ (old batch $\sigma=0.064$), significant convergence of fluctuations but mean below 0.72 - recommend Flow/EM increase by 1-3% to bring mean to 0.73-0.74 to improve C_{pk} (C_{pl}).

Width

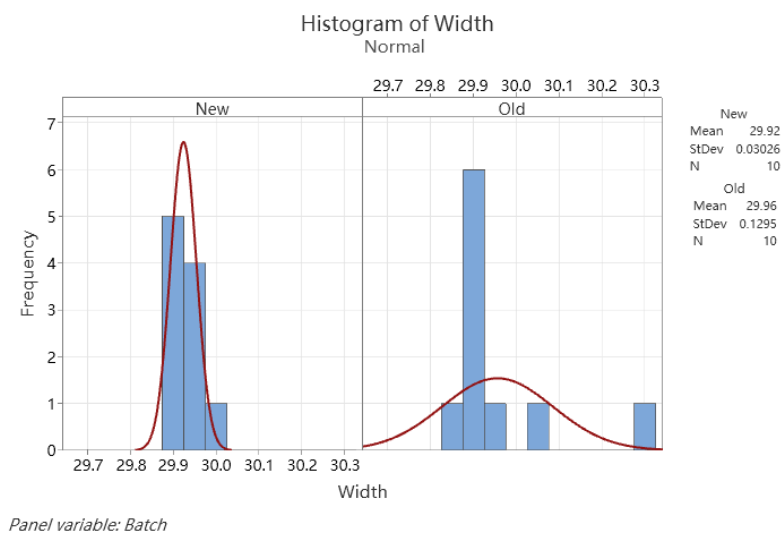


Figure 6: Histogram of Width

Width: new batch mean 29.92, σ 0.030 (old batch σ 0.130), fluctuations converge significantly and stabilise in the centre of the specification; capacity significantly better than old batch.

Length

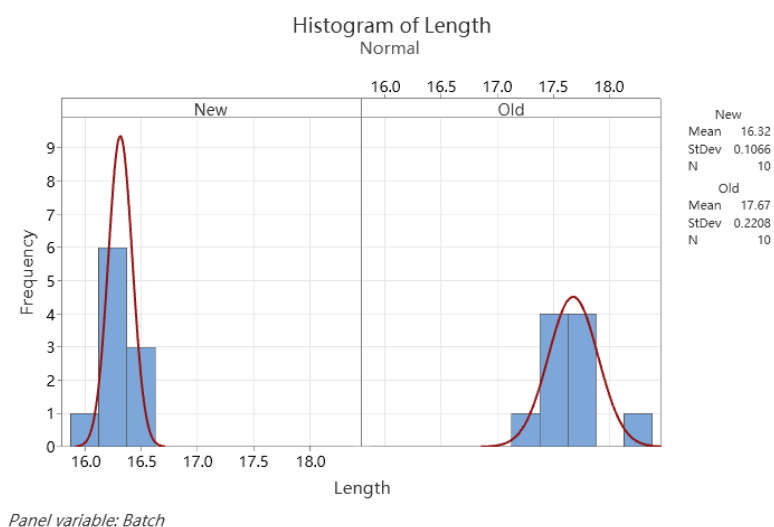


Figure 7: Histogram of Length

New averages 16.32, σ 0.1066; Old averages 17.67, σ 0.2208, suggesting that the new batches bring the length of the batch much closer to 16 from 17-18, halving the fluctuations, but remaining higher than USL=15.5 overall.

Three-size box line diagram

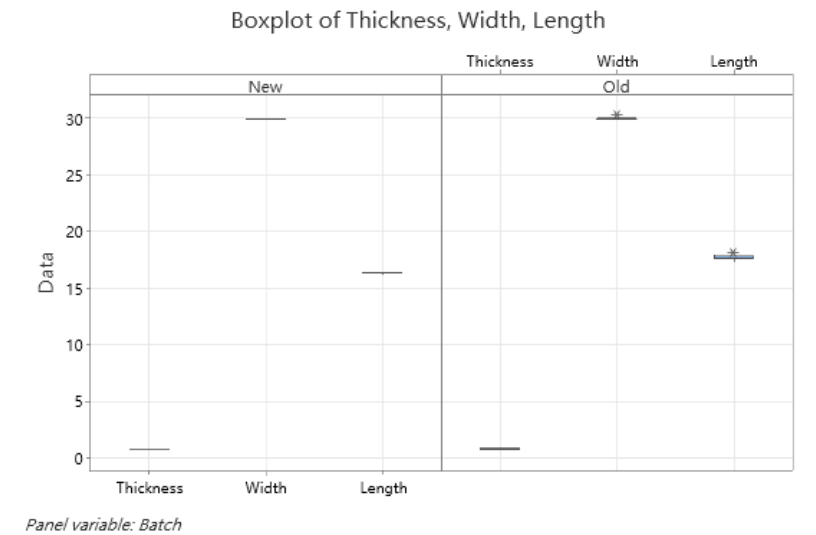


Figure 8: Histogram of Thickness, Width and Length

The three key performance indicators for the new batch reveal: The box length has been significantly reduced (with fluctuations tending towards convergence), width is now largely centred and stable; however, the average thickness has decreased slightly, while length remains biased towards and partially exceeds the USL limit.

Capacity analysis (C_p/C_{pk})

Thickness:

Old Hinges: The thickness mean is high, close to USL=0.82, and fluctuates widely, with $C_p \approx 0.52$ and $C_{pk} \approx 0.27$ (constrained by the upper side), and about 20% would be expected to fall outside of the USL by a normal fit - no wonder there are super-thick pieces in the old batch.

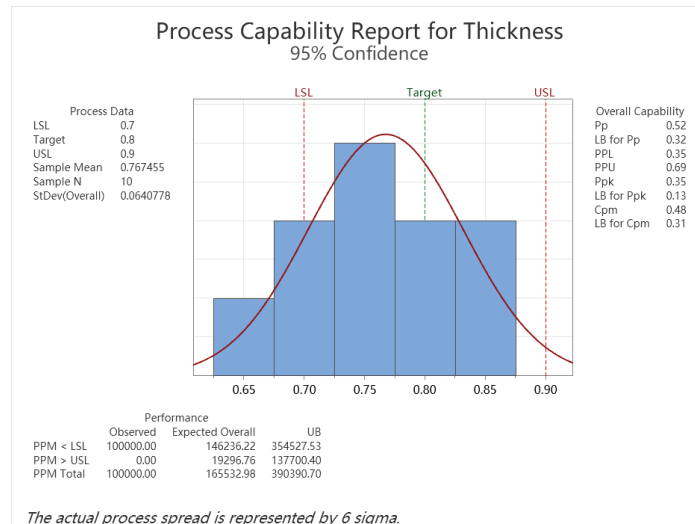


Figure 9: Process Capability Report for Thickness (Old Hinges)

New Hinges: New batch thickness mean 0.706 mm, overall σ 0.0104, very capable ($P_p \approx 3.21$, $P_{pk} \approx 2.76$, away from LSL = 0.7), but a little low relative to Target 0.8; recommend 1-3% increase in Flow/EM to bring mean to 0.73-0.74, to improve centrality (C_{pm}) and to provide a base for life improvement. It is recommended that Flow/EM be adjusted upwards by 1-3% to bring the mean to 0.73-0.74 to improve centrality (C_{pm}) and to provide a base for lifetime improvement.

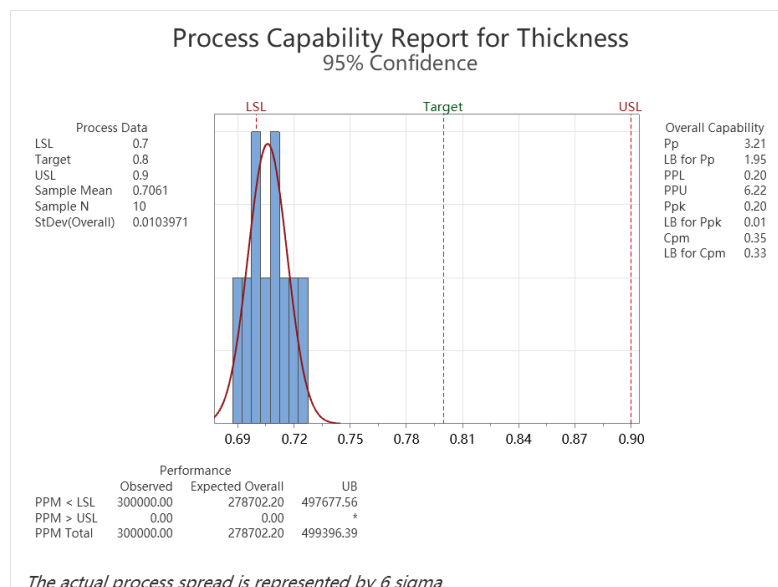


Figure 9: Process Capability Report for Thickness (New Hinges)

Width:

Old Hinges: Mean 29.956, σ 0.129, $P_p \approx 2.57$, $P_{pk} \approx 2.46$ (95% CI 1.31-3.62); the distribution is close to 30 and far from the tolerance boundaries, with almost 0 PPM overshoots - the capacity is very ample and essentially centred.

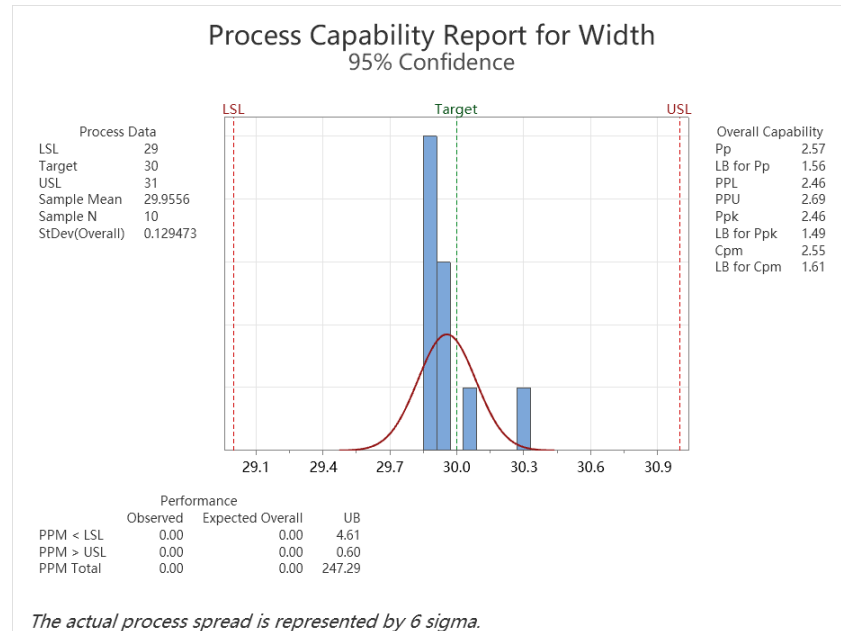


Figure 10: Process Capability Report for Width (Old Hinges)

New Hinges: Mean 29.924 mm, overall σ 0.0303 mm (n=10), $P_p \approx 11.0$, $P_{pk} \approx 10.2$ (95% CI 5.47-14.89), almost 0 PPM overshoot; widths are highly stable, slightly below 30 but well centred ($C_{pm} \approx 4.10$).

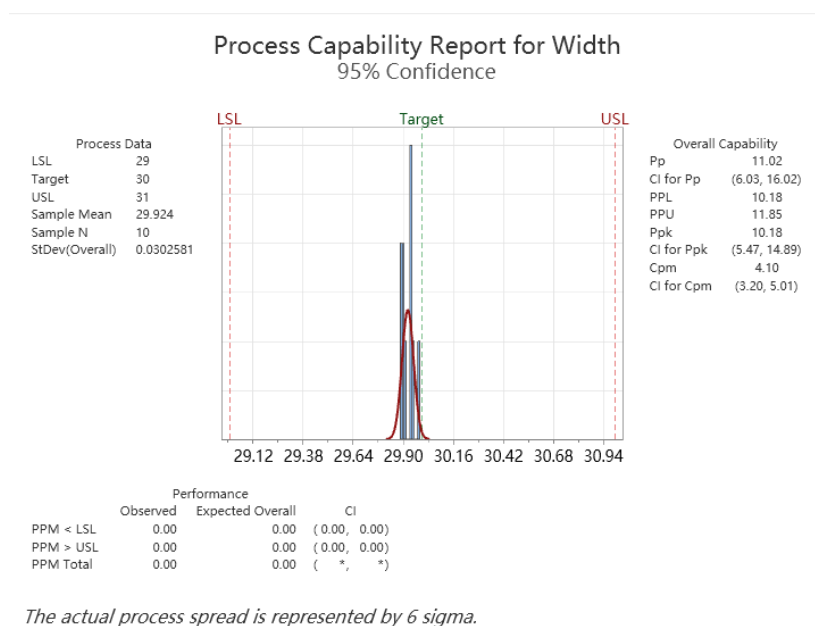


Figure 11: Process Capability Report for Width (New Hinges)

Length:

Old Hinges: The systematic over-ceiling of old batch lengths is the main cause of capacity failure, although the fluctuations (σ) are not considered large; any C_p will be crushed by the eccentricity into a negative C_{pk} .

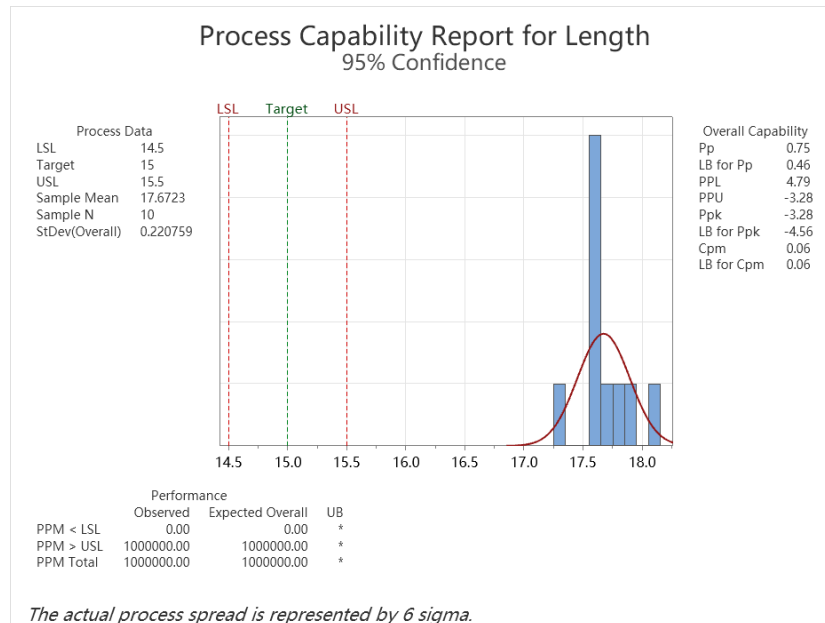


Figure 12: Process Capability Report for Length (Old Hinges)

New Hinges: The new batch keeps the fluctuations small, but the mean is still higher than $USL=16$, so the C_{pk} is negative - this is not a case of "poor capacity", but rather a case of systematic bias leading to "overall overshooting".

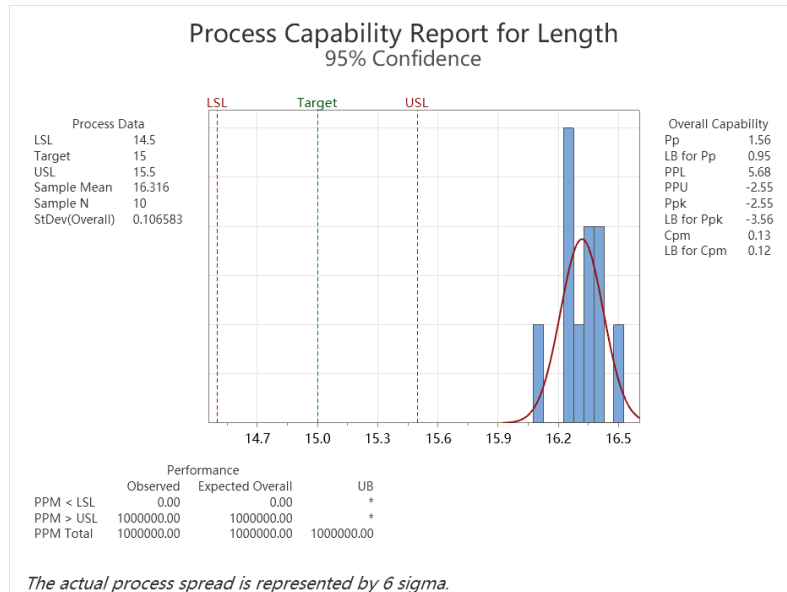


Figure 13: Process Capability Report for Length (New Hinges)

Part Quality

Thickness_ok	Width_OK	Length_OK	PartQuality
1	1	0	0.666667
1	1	0	0.666667
0	1	0	0.333333
1	1	0	0.666667
1	1	0	0.666667
1	1	0	0.666667
0	1	0	0.333333
1	1	0	0.666667
1	1	0	0.666667
1	1	0	0.666667
1	1	0	0.666667
1	1	0	0.666667
1	1	0	0.666667
1	1	0	0.666667
1	1	0	0.666667
1	1	0	0.666667
1	1	0	0.666667
1	1	0	0.666667
1	1	0	0.666667
1	1	0	0.666667

Figure 14: Table for Part Quality (1 means ok, 0 means Bad)

Statistics

Variable	Batch	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3
PartQuality	New	10	0	0.66667	0.000000	0.000000	0.66667	0.66667	0.66667	0.66667
	Old	10	0	0.6000	0.0444	0.1405	0.3333	0.5833	0.6667	0.6667

Variable	Batch	Maximum
PartQuality	New	0.66667
	Old	0.6667

Figure 15: Statistics for Part Quality

Average quality 0.6000→0.6667; Improvement stems from convergence in thickness/width and overall pass rate, with length remaining the weak point.

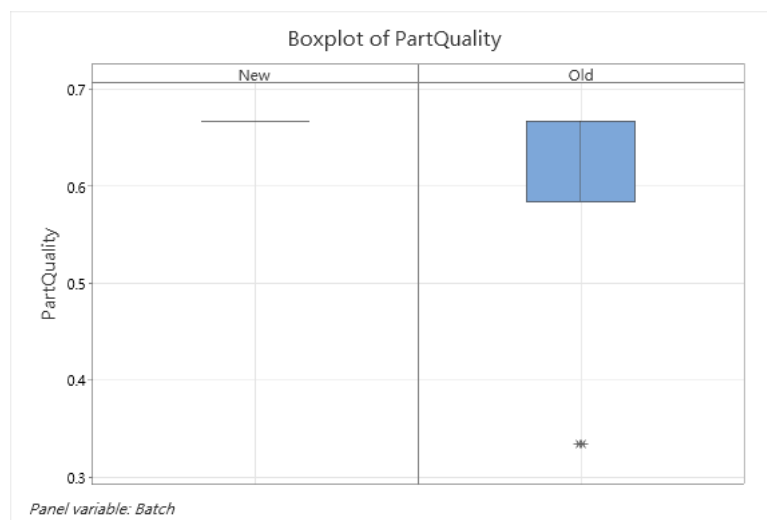


Figure 16: Boxplot of Part Quality

Old vs New Size Difference

Thickness: The difference in thickness between the two batches is significant, with New being significantly thinner; in combination with the previous capacity plot, New is also less volatile.

Method

η_1 : median of Thickness_Old

η_2 : median of Thickness_New

Difference: $\eta_1 - \eta_2$

Descriptive Statistics

Sample	N	Median
Thickness_Old	10	0.753278
Thickness_New	10	0.707000

Estimation for Difference

Difference	CI for Difference	Achieved Confidence
0.0462778	(0.0148889, 0.105667)	95.48%

Figure 17: Mann-Whitney: Thickness_Old, Thickness_New 1

Test

Null hypothesis $H_0: \eta_1 - \eta_2 = 0$

Alternative hypothesis $H_1: \eta_1 - \eta_2 \neq 0$

Method	W-Value	P-Value
Not adjusted for ties	140.00	0.009
Adjusted for ties	140.00	0.009

Figure 18: Mann-Whitney: Thickness_Old, Thickness_New 2

Width: No significant difference between the two batches (Mann-Whitney $p=0.569$), with a median difference of -0.010 mm (95% CI -0.050 to 0.051). Both remain stable at centres 29–31, requiring no adjustment. When subsequently enabling elephant foot/XY compensation to correct length, note that width should still be maintained close to 30.

Mann-Whitney: Width_Old, Width_New

Method

η_1 : median of Width_Old
 η_2 : median of Width_New
Difference: $\eta_1 - \eta_2$

Descriptive Statistics

Sample	N	Median
Width_Old	10	29.9039
Width_New	10	29.9250

Estimation for Difference

Difference	CI for Difference	Achieved Confidence
-0.0100000	(-0.0500000, 0.0511111)	95.48%

Test

Null hypothesis $H_0: \eta_1 - \eta_2 = 0$
Alternative hypothesis $H_1: \eta_1 - \eta_2 \neq 0$

Method	W-Value	P-Value
Not adjusted for ties	97.00	0.571
Adjusted for ties	97.00	0.569

Figure 19: Mann-Whitney: Mann-Whitney: Width_Old, Width_New

Length: Significant difference ($p < 0.001$); median difference +1.322 mm (95% CI 1.213-1.501) → New significantly shorter but mean still above USL=16. Response: Elephant-Foot Compensation 0.15-0.25 mm + XY Size Compensation -0.2% to -0.4% adjusted in small steps, and width reviewed each time.

Mann-Whitney: Length_Old, Length_New

Method

η_1 : median of Length_Old
 η_2 : median of Length_New
Difference: $\eta_1 - \eta_2$

Descriptive Statistics

Sample	N	Median
Length_Old	10	17.6161
Length_New	10	16.3250

Estimation for Difference

Difference	CI for Difference	Achieved Confidence
1.32222	(1.21333, 1.50111)	95.48%

Test

Null hypothesis $H_0: \eta_1 - \eta_2 = 0$
Alternative hypothesis $H_1: \eta_1 - \eta_2 \neq 0$

Method	W-Value	P-Value
Not adjusted for ties	155.00	0.000
Adjusted for ties	155.00	0.000

Figure 20: Mann-Whitney: Length_Old, Length_New

Non-KPI Comparison (Old vs New)

Number	Non-KPI phenomena (Old → New)	Evidence/data support	Possible causes (mechanisms)	Improvement actions (priority)	Expected Results & Verification Methods
F1	Primary outgrowth/XY bias → still present (indirectly reflected by KPIs)	New Length mean 16.316 mm, $p_{PK} = -0.99$ (super USL=16); Old 17.67	The first layer of "elephant foot" is slightly enlarged with XY dimensions.	P0: PrusaSlicer Prism Compensation 0.15–0.25 mm; P0: XY Dimensional Compensation -0.2% to -0.4% in small increments; Re-measure after printing 10 pieces each time.	Length back to ≤ 16 ; PartQuality → 1.00; Review Width still at 29-31

		mm (capacity plot vs. histogram)			
F2	More stable surface consistency (more even lamination)	σ : thickness 0.0104 (Old 0.0641), length 0.1066 (Old 0.2208)	Layer height 0.15 mm, dry consumables, air- cooled for greater stability	P1: Maintain 0.15 mm QUALITY; continue drying consumables, fan 50- 70 per cent.	Improved surface roughness and layer uniformity; visual inspection & sidelight photo archiving.
F3	Lighter seams/splicing marks (expected/if additional walls have been added)	New Width vs. thickness Cp/Cpk Higher and more centrally distributed	Reduced circumferential stresses and load sharing by wall number 4; more stable peripheral line overlap	P1: number of walls 2 → 4 (already implemented, then kept); joints set to "random/hidden to the back"	Cleaner edges, no protruding seams; edge caliper sampling
F4	Warping/drawing mitigation (if observed)	Cycles with improved dimensiona l stability for a more stable	First layer attachment/cooling/ret raction optimisation	P2: first layer temperature and air volume levelling; back pumping 0.6-1.0 mm, speed 20-30 mm/s	Decrease in warpage/dra wing defect rate; defect point count table

		process.			
F5	Print consistency ↑ (less dispersion within a batch)	New Part Quality = 0.667 (all equal); Old Mean 0.600, SD 0.1405	Better process control and measurement consistency	P1: Maintain same roll/environment/para meters; same machine, same batch printing	Process fluctuations continue to converge; each batch of PQ averages is monitored with SD
F6	Traceability/measurement processes (normalisation)	H1-H10 unique identification; CLP- 01 caliper calibration ±0.01 mm; average of three thickness measurements	Consistency of gauges and methods to avoid mix-ups/missing notes	P1: Continuation of the process; dimensional checking error ≤0.02 mm; complete record sheet	Data are complete and traceable; zero defects in spot checks
F7	Functional feel (hinge movement) (recommended recording)	Not objectively quantifiable	Size/first floor/clearance co- influence	P2: Create a 3-point subjective scale (smooth/slightly tight/not smooth); leave 0.1-0.2 mm clearance in shaft/hole if	Functional consistency is well documented; correlation analysis with length/thickn

				necessary.	ess
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Does the thickness deviate from 0.72?

Old Hinges: Thickness is "more stable but low" on New and "significantly high" on Old; next round of Flow/EM +1-3% with established EF/XY dimensional compensation (to correct for Length). The goal is to have thickness centred, length back in line, and PartQuality close to 1.00.

One-Sample T: Thickness

Descriptive Statistics

N	Mean	StDev	SE Mean	95% Lower Bound for μ
10	0.7675	0.0641	0.0203	0.7303

μ : population mean of Thickness

Test

Null hypothesis $H_0: \mu = 0.72$
Alternative hypothesis $H_1: \mu > 0.72$

T-Value	P-Value
2.34	0.022

Figure 21: Mann-Whitney: One-Sample T: Thickness Old

New Hinges: New's mean thickness is significantly lower relative to Target = 0.72 mm (1-Sample t: $t = -4.23$, $p = 0.001$; 95% upper bound $\mu \leq 0.712$). Recommendation: Adjust Flow/EM upwards by 1-3% to bring the mean to 0.73-0.74, to improve centrality (C_{pm}) and provide a base for lifetime improvement.

One-Sample T: Thickness

Descriptive Statistics

N	Mean	StDev	SE Mean	95% Upper Bound for μ
10	0.70610	0.01040	0.00329	0.71213

μ : population mean of Thickness

Test

Null hypothesis $H_0: \mu = 0.72$
Alternative hypothesis $H_1: \mu < 0.72$

T-Value	P-Value
-4.23	0.001

Figure 22: Mann-Whitney: One-Sample T: Thickness New

Q2. Reliability Life Assessment (Old vs New): Testing, Differences and Improvements

Lifetime test and breakdown summary

New group survival duration concentrated between 15–40 cycles (median 27); old group exhibited early loss plus long tail (median 73). Log-rank test $p=0.051$, Wilcoxon test $p=0.438$; differences primarily in the tail, with overlap in the initial segment. Interquartile range: new 13 vs old 129 (high variability in old group).

Distribution Analysis: Cycles by Batch

Variable: Cycles
Batch = New

Censoring

Censoring Information Count	
Uncensored value	10

Nonparametric Estimates

Characteristics of Variable

		Standard Error		95.0% Normal CI			
Mean(MTTF)		Lower	Upper	Q1	Median	Q3	IQR
28.2	2.53333	23.2348	33.1652	23	27	36	13

Kaplan-Meier Estimates

Time	Number at Risk	Number Failed	Survival Probability	Standard Error	95.0% Normal CI	
					Lower	Upper
15	10	1	0.9	0.094868	0.714061	1.00000
23	9	2	0.7	0.144914	0.415974	0.98403
24	7	1	0.6	0.154919	0.296364	0.90364
27	6	2	0.4	0.154919	0.096364	0.70364
28	4	1	0.3	0.144914	0.015974	0.58403
36	3	1	0.2	0.126491	0.000000	0.44792
37	2	1	0.1	0.094868	0.000000	0.28594
42	1	1	0.0	0.000000	0.000000	0.00000

Figure 23: Kaplan-Meier Survival Estimates and Statistics

Distribution Analysis: Cycles by Batch

Variable: Cycles
Batch = Old

Censoring

Censoring Information Count	
Uncensored value	10

Nonparametric Estimates

Characteristics of Variable

		Standard Error		95.0% Normal CI			
Mean(MTTF)		Lower	Upper	Q1	Median	Q3	IQR
97.8	36.6336	25.9995	169.600	1	73	130	129

Kaplan-Meier Estimates

Time	Number at Risk	Number Failed	Survival Probability	Standard Error	95.0% Normal CI	
					Lower	Upper
1	10	3	0.7	0.144914	0.415974	0.984026
3	7	1	0.6	0.154919	0.296364	0.903636
73	6	1	0.5	0.158114	0.190102	0.809898
75	5	1	0.4	0.154919	0.096364	0.703636
104	4	1	0.3	0.144914	0.015974	0.584026
130	3	1	0.2	0.126491	0.000000	0.447918
252	2	1	0.1	0.094868	0.000000	0.285939
338	1	1	0.0	0.000000	0.000000	0.000000

Figure 24: Kaplan-Meier Lifespan Summary (Old Lot)

Distribution Analysis: Cycles by Batch

Comparison of Survival Curves

Test Statistics

Method	Chi-Square	DF	P-Value
Log-Rank	3.81636	1	0.051
Wilcoxon	0.60046	1	0.438

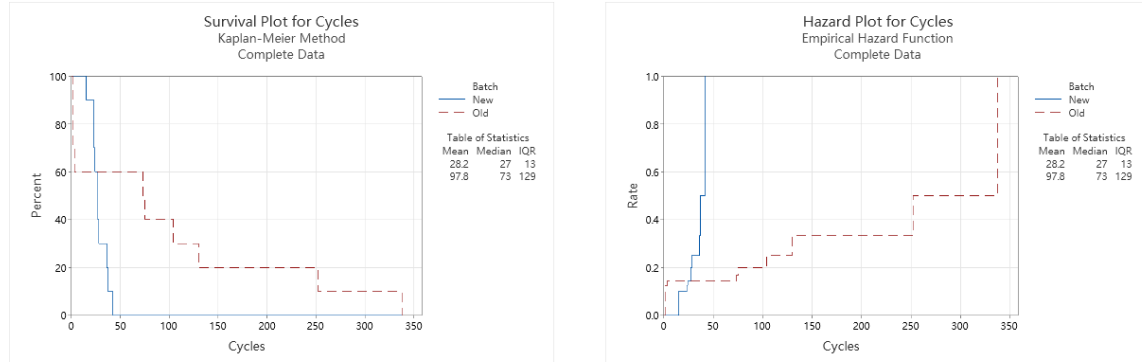


Figure 25: Survival vs Risk Curve (Old vs New)

Test Fixture and Measurement System Adequacy Assessment

Checkpoint	reach a verdict	Evidence/Remarks
Stroke and speed can be reproduced	✓	Beat/stopwatch checking; each time from the same starting point
Coaxial and Boundary	✓	As illustrated in the diagram; do not cut to EF.
Consistency of counts with judgements	✓	Counter = video frame; Failure = failure to return/penetrating cracks
Randomisation and consistency	✓	Old/New mixed rows; same operator/environment
Traceability and measurement	✓	H1-H10 uniquely labelled; CLP-01 ±0.01 mm; thickness triple-tested for uniformity

Computational process

$$\sigma_g = \frac{0.01}{\text{sqrt}(12)} = 0.00289 \text{ mm}$$

$$6\sigma_y = 0.00289 * 6 = 0.01732 \text{ mm}$$

$$\frac{P}{T} = \frac{6\sigma_y}{T}$$

Thickness: $0.01732/0.2 = 8.66\%$

Width: $0.01732/2 = 0.87\%$

Length: $0.01732/2 = 0.87\%$

Results matrix

Dimensions	Tolerance T (mm)	Resolution R (mm)	σ_g	$6\sigma_y$	P/T	Reach a verdict
Thickness	0.2	0.01	0.00289	0.01732	8.66%	Good (≤ 10 per cent)
Width	2	0.01	0.00289	0.01732	0.87%	Talented
Length	2	0.01	0.00289	0.01732	0.87%	Talented

The "cause" of life variance is related to size/process.

The Mann-Whitney test showed that the difference in the distribution of lifespans between Old and New was not significant ($W=115$, $p=0.472$); although the median sample size was higher for Old (74 vs 27), the 95% CI of the difference (-24, 106) spanned 0, suggesting that with the current sample size and fluctuations it was not possible to reject the null hypothesis.

Mann-Whitney: Cycles_Old, Cycles_New

Method

η_1 : median of Cycles_Old
 η_2 : median of Cycles_New
Difference: $\eta_1 - \eta_2$

Descriptive Statistics

Sample	N	Median
Cycles_Old	10	74
Cycles_New	10	27

Estimation for Difference

Difference	CI for Difference	Achieved Confidence
47.5	(-24, 106)	95.48%

Test

Null hypothesis $H_0: \eta_1 - \eta_2 = 0$
Alternative hypothesis $H_1: \eta_1 - \eta_2 \neq 0$

Method	W-Value	P-Value
Not adjusted for ties	115.00	0.473
Adjusted for ties	115.00	0.472

Figure 26: Mann-Whitney: Cycles_Old, Cycles_New

Old Hinges: In older batches, lifespan showed little correlation with thickness, width, or length; the real determining factor was processing quality. The pronounced inverse relationship between thickness and width (reflecting trade-offs between flow, XY compensation, and elephant footing) indicates that extending lifespan requires first tightening processes like interlayer adhesion and first-layer/seam bonding.

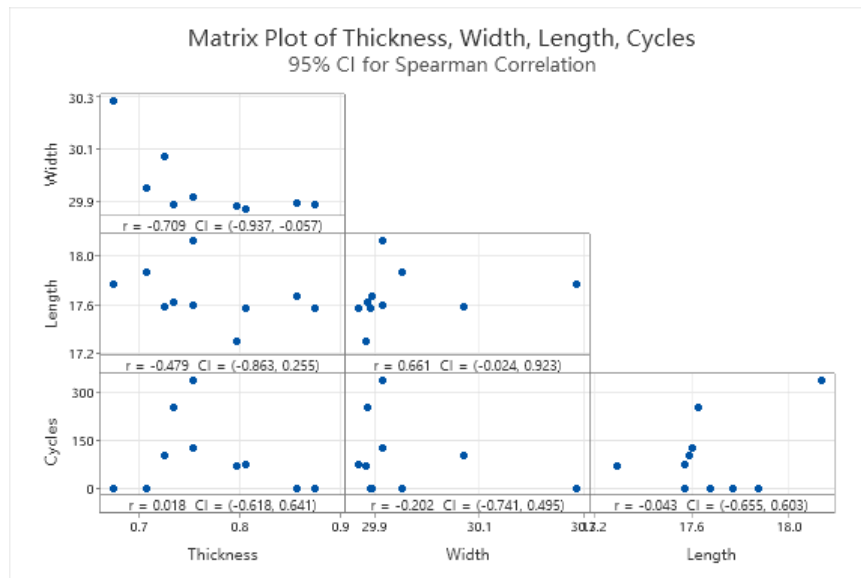


Figure 27: Old batch: Spearman's correlation matrix of thickness/width/length versus life (with 95% confidence intervals).

Method

Correlation type Spearman
Number of rows used 10

Correlations

	Thickness	Width	Length
Width	-0.709		
Length	-0.479	0.661	
Cycles	0.018	-0.202	-0.043

Figure 28: Old batch: Spearman's correlation coefficients for thickness/width/length vs. life (n=10)

New Hinges: The correlation between life and thickness/width/length in the New batch is not significant, suggesting that geometry is not the main cause, and that improving life should focus on the process (flow/cooling/seams/first layer, etc.), while thickness-width and width-length still vaguely show a tendency to trade-off between flow and XY compensation.

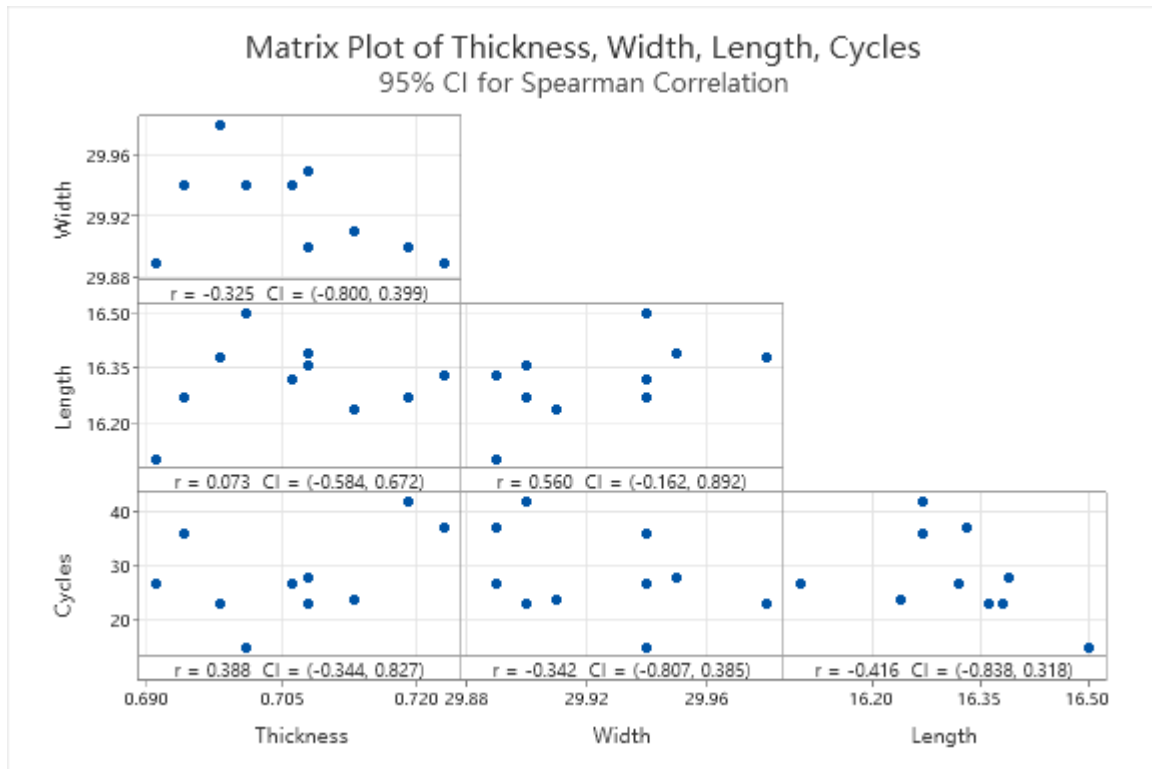


Figure 29: New batch: Spearman's correlation matrix of thickness/width/length versus life (with 95% confidence intervals).

Method

Correlation type Spearman
Number of rows used 10

Correlations

	Thickness	Width	Length
Width	-0.325		
Length	0.073	0.560	
Cycles	0.388	-0.342	-0.416

Figure 30: New batch: Spearman's correlation coefficients for thickness/width/length vs. life (n=10)

$\rho(\text{Spearman})$	Old	New
Cycles vs Thickness	0.018	0.388
Cycles vs Width	-0.202	-0.342
Cycles vs Length	-0.043	-0.416

The lifespan of New remains insignificantly linked to the ruler; improving its fate still hinges on craftsmanship—tightening the screws on interlayer bonding, the first layer and joints, and cooling/drying processes will finally see the curve rise.

Failure modes and lifetime mechanisms: from observation to causation

New Hinges' early failures are mainly triggered by "first layer/sticky bed (FM3)" (N=9, Median ≈ 27 times), while Old's "Stable_NoNotch" lasts significantly longer (Median ≈ 252 times); seams (FM2) are under-sampled (N=1 for reference only). ≈ 252 times); seams (FM2) are under-sampled (N=1) for information only.

Rows: FM_suspect		Columns: Batch		
		New	Old	All
FM2_Seam		42.00	*	42.00
		42.00	*	42.00
		42	*	42
		42	*	42
		*	*	*
		1	0	1
		1	0	1
FM3_FirstLayer		26.67	*	26.67
		27.00	*	27.00
		15	*	15
		37	*	37
		6.76	*	6.76
		9	0	9
		9	0	9
Stable_NoNotch		*	240.00	240.00
		*	252.00	252.00
		*	130	130
		*	338	338
		*	104.52	104.52
		0	3	3
		0	3	3
Unknown		*	36.86	36.86
		*	3.00	3.00
		*	1	1
		*	104	104
		*	45.23	45.23
		0	7	7
		0	7	7
All		28.20	97.80	63.00
		27.00	74.00	27.50
		15	1	1
		42	338	338
		8.01	115.85	87.53
		10	10	20
		10	10	20
Cell Contents				
Cycles : Mean				
Cycles : Median				
Cycles : Minimum				
Cycles : Maximum				
Cycles : Standard deviation				
Cycles : Nonmissing				
Count				

Figure 31: Lifespan statistics (Cycles)

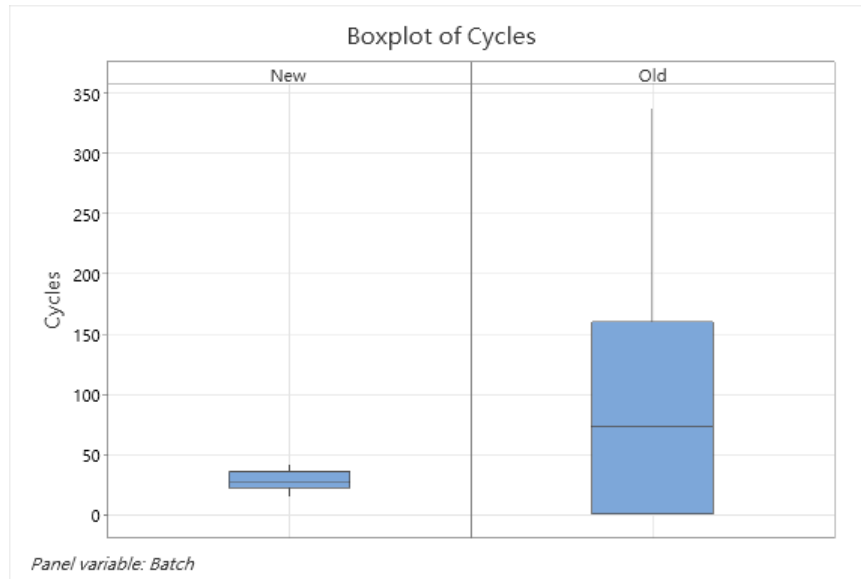


Figure 32: Lifespan (Cycles) box plot: surface by batch (New vs Old)

Processes to improve longevity

Number/Action	Parameters (setting recommendation s)	Rationale / Mechanism	Intended effect	Validation and quantification
P0 Flow/EM Upward Adjustment +1~3%	Flow/EM first +2% prints 10 pieces; fine tune to +1~3% if necessary	The low mean thickness value will weaken the interlayer bonding, first pull back the mean thickness value to 0.72±.	Mean thickness 0.706 → 0.73- 0.74; C_{pm} ↑, P_{pk} ↑; interlayer bonding more stable	Do Capability (Thickness: LSL 0.62 / USL 0.82) juxtaposition plots; record increases in Pp/Ppk, Cpm; median lifetime rise
P1 XY/first floor compensation, eliminating elephant trunk opening	Elephant-Foot 0.15-0.25 mm; XY Compensation - 0.2% to -0.4%;	"Elephant trunk/steps" = Typical early notch that triggers early	Early hazard decreases, left end of KM survival curve increases	Comparison of 10 pieces: record KM curve (By=Batch) to see improvement at left end; photo of

	First Floor Height 0.20 mm; First Floor Flow -5%	fracture		failed first floor section
P2 Joint control (Seam control)	Seam = Random or "hidden to the unstressed side"; Seam Painting preferred; small outer ring 25-30 mm/s	New Hinges' FM2 (joint cracking) is a major early failure	Reduced seam protrusion; median lifetime ↑, dispersion ↓	Cycles vs Thickness/Width/Len gth (By=Batch) Scatter convergence; microscopic/magnifie d photo comparison at seams
P3 Number of Walls & Floor Height (Intensity Baseline)	Outer wall 4 perimeters; layer height 0.12 mm (first 2 layers still 0.20); Z-direction fixation	Multiple thin layers of wall → longer crack paths, larger bond area	High peripheral endurance ↑; IQR narrowed, right tail longer/more stable	Boxplot: Cycles (By=Batch) vs. IQR; Decreased proportion of interlayer stripping in failure mode
P4 Temperature/Cooling/ Air	Nozzle temperature +5 to 10 °C; Fan 50– 70% (gradually increase to full speed after the third layer); Reduce speed by 10–20% at small cross-sections.	Interlayer soldering requires a window of heat, too cold or too fast will result in "false soldering".	Reduced FM1 (interlayer delamination), denser fracture	Decreased percentage of FM1 in failure mode Pareto; fracture photo control
P5 Material Drying and Storage	55-60 °C × 4-6 h drying; Ziploc	Melt with water can cause	Reduced whitening/stomat	Comparative dry/wet material

	bag + desiccant; change to new material for control	porosity/whitenin g, weakening the life span	al sections; Cycles median improved in tandem with IQR	slices/photographs; box plot variations; batch and feed ledgers
P6 Measurement and Process (Traceability)	Caliper calibration ± 0.01 mm; H1-H10 uniquely labelled; Thickness taken as a 3-point average;	Stabilisation of capabilities by turning "random good and bad" into a controlled process	P_p/P_{pk} , $C_{pm} \geq$ 2; PartQuality median shifted up, box narrowing	Save Capability reports, process pictures, gauge calibration certificates; PartQuality statistics/charts.

Parametric Life Modelling

Old batch life distribution fitting (parametric method) + 4 types of function plots (PDF/CDF/R(t)/Hazard)

Sports event	Numerical / Conclusion	Note
Fit a distribution (math.)	Weibull (2 parameters)	Similar to the Loglogistic fit; the three-parameter model was unstable under this sample and was not used
Goodness of fit	AD = 1.922	Superior in candidate distributions; fits well with probability map points
Shape parameter β	0.550 (95%CI 0.326 – 0.925)	$\beta < 1 \Rightarrow$ Hazard rate decreases over time, with early failure dominance, followed by a long tail
Scale parameter η	64.686 (95%CI 19.749 – 211.869)	Characteristic life
MTTF	110.26 cycles (95%CI 32.90 –	Consistent with the sample

	369.50)	median of 33.20 cycles, "long-tailed, large dispersion".
t_{10} (10 per cent time to failure)	≈ 10.18 cycles (95% CI: very low lower limit, upper limit ≈ 16.76)	Exported as Minitab "Distribution Overview + Quadruple Chart", can be directly inserted.
Chart (4 classes of functions)	PDF / CDF / R(t) / h(t)	Exported as Minitab 'Distribution Overview + Quadrant Plot'
Concluding remarks	Early Defect Control is the Key to Enhanced Longevity	Priority convergence of first layer bonding/seam/cooling and material drying; MTTF and dispersion will improve in tandem with the reduction of early failures

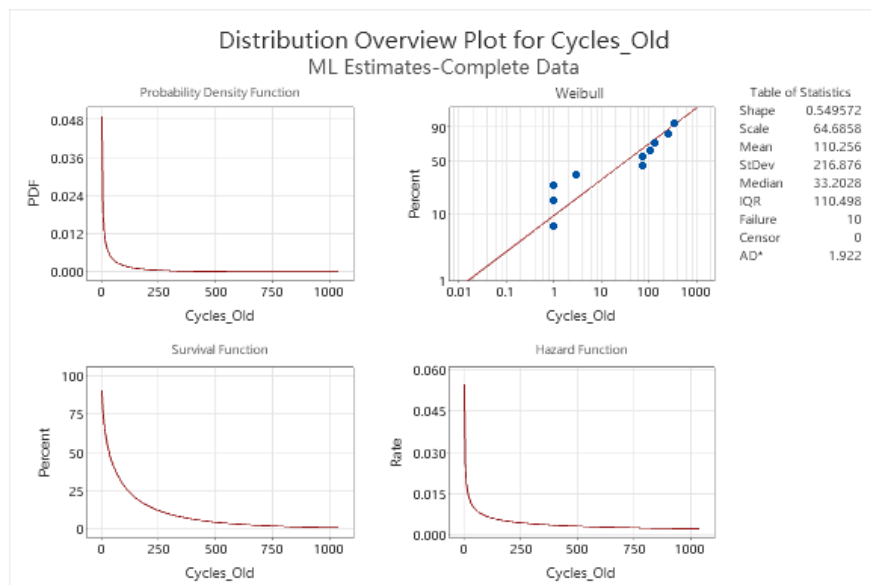


Figure 33: Distribution Overview – Old Hinges (Weibull): PDF, Probability, Survival, Hazard

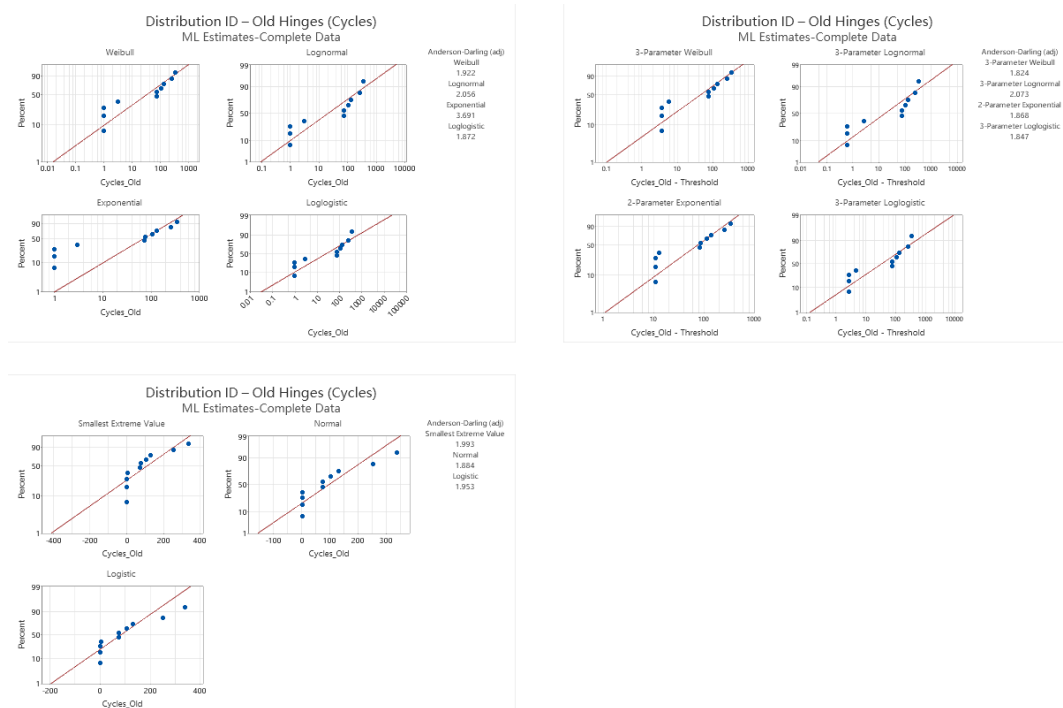


Figure 34: Old Batch Life Distribution Identification (Distribution ID) - Multi Model Comparison

Interval estimation: MTTF vs. 10% downtime

Sports event	Results/Description
Fit a distribution (math.)	Weibull (two-parameter)
Goodness of fit	AD = 1.922 (preferred among alternative models; the 3-parameter model was unstable and not adopted)
Shape parameter β	0.5496 , 95% CI: 0.326–0.925
Scale parameter η	64.69 , 95% CI: 19.75–211.87
MTTF	110.26 , 95% CI: 32.90–369.46
10% Time to failure t10	$\approx$ 1.08 , 95% CI: [0.16, 6.10] (Percentiles Table 10% row)
Account	$\beta < 1 \rightarrow$ decreasing hazard: early failures dominate, surviving samples gradually stabilise.

Distribution Analysis: Cycles_Old

Variable: Cycles_Old

Censoring

Censoring Information Count	
Uncensored value	10

Estimation Method: Maximum Likelihood

Distribution: Weibull

Parameter Estimates

Parameter	Estimate	Standard Error	95.0% Normal CI	
			Lower	Upper
Shape	0.549572	0.146079	0.326415	0.925292
Scale	64.6858	39.1563	19.7493	211.869

Log-Likelihood = -52.671

Goodness-of-Fit

Anderson-Darling (Adjusted)	
	1.922

Characteristics of Distribution

	Estimate	Standard Error	95.0% Normal CI	
			Lower	Upper
Mean(MTTF)	110.256	68.0244	32.9029	369.459
Standard Deviation	216.876	175.959	44.2180	1063.71
Median	33.2028	22.6289	8.73075	126.270
First Quartile(Q1)	6.70266	6.55380	0.986165	45.5559
Third Quartile(Q3)	117.200	67.5295	37.8858	362.561
Interquartile Range(IQR)	110.498	63.6063	35.7579	341.456

Table of Percentiles

Percent	Percentile	Standard Error	95.0% Normal CI	
			Lower	Upper
1	0.0149823	0.0371642	0.0001159	1.93649
2	0.0533745	0.114941	0.0007840	3.63393
3	0.112664	0.221141	0.0024044	5.27904
4	0.191952	0.350870	0.0053368	6.90406
5	0.290828	0.501221	0.0099232	8.52353
6	0.409125	0.670227	0.0164976	10.1459
7	0.546829	0.856459	0.0253907	11.7768
8	0.704027	1.05883	0.0369329	13.4204
9	0.880888	1.27650	0.0514566	15.0800
10	1.07764	1.50879	0.0692978	16.7583
20	4.22176	4.55763	0.508830	35.0279
30	9.91116	8.87862	1.71238	57.3652
40	19.0538	14.6864	4.20617	86.3133
50	33.2028	22.6289	8.73075	126.270
60	55.1728	34.1941	16.3750	185.896
70	90.6774	52.9306	28.8823	284.686
80	153.771	88.7184	49.6329	476.405
90	295.051	181.282	88.4929	983.749
91	320.076	199.192	94.5205	1083.88
92	349.133	220.474	101.267	1203.69
93	383.445	246.231	108.918	1349.92
94	424.848	278.152	117.745	1532.93
95	476.270	318.972	128.168	1769.82
96	542.784	373.521	140.885	2091.17
97	634.270	451.416	157.205	2559.07
98	774.002	575.924	180.042	3327.44
99	1041.48	829.486	218.631	4961.23

Figure 36: Old Batch Life Distribution (Weibull) - Parameter Estimation and Life Statistics

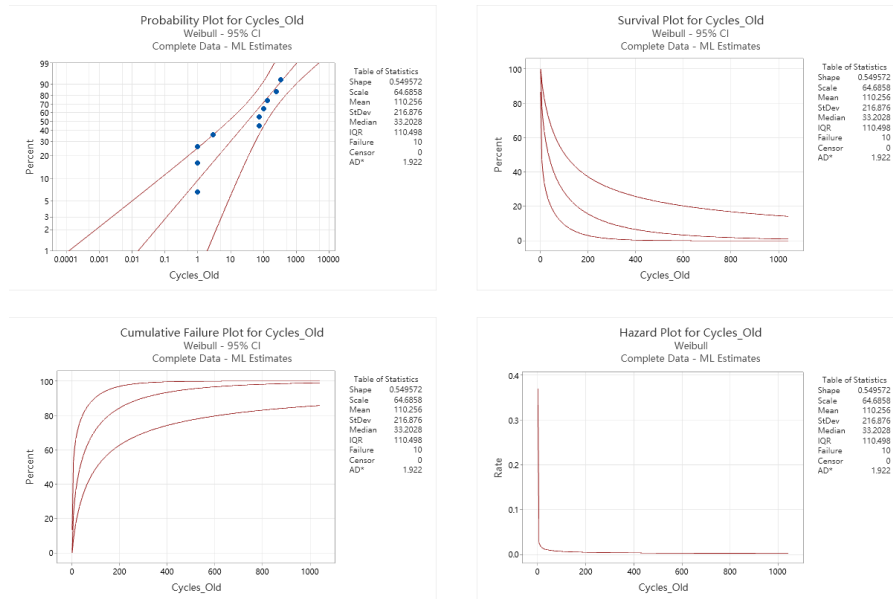


Figure 36: Old Batch Life Distribution (Weibull) Four Plots: Probability Plot - Survival - Cumulative Failure - Risk Ratio

Reliability improvement assessment of new batch hinges

New batch-by-batch life and failure records

Part	Cycles (New)	Censor	FM_suspect	Thickness_OK (1 means pass)	Width_OK (1 means pass)	Length_OK (0 means not pass)
1	28	0	FM3_FirstLayer	1	1	0
2	15	0	FM3_FirstLayer	1	1	0
3	37	0	FM3_FirstLayer	1	1	0
4	27	0	FM3_FirstLayer	1	1	0
5	23	0	FM3_FirstLayer	1	1	00
6	23	0	FM3_FirstLayer	1	1	0
7	42	0	FM2_Seam	1	1	0
8	36	0	FM3_FirstLayer	1	1	0
9	27	0	FM3_FirstLayer	1	1	0
10	24	0	FM3_FirstLayer	1	1	0

↓	C1-T	C2	C3	C4	C5	C6	C7	C8-T	C9	C10	C11	C12	C13	C14
	Batch	Part	Thickness	Width	Length	Cycles	Censor	FM_suspect	Thickness_ok	Width_OK	Length_OK	PartQuality	Cycles_New	Cycles_Old
1	Old	1	0.796667	29.8833	17.2967	73	0	Unknown	1	1	0	0.666667	28	73
2	Old	2	0.733889	29.8867	17.6278	252	0	Stable_NoNotch	1	1	0	0.666667	15	252
3	Old	3	0.854889	29.8944	17.6778	1	0	Unknown	0	1	0	0.333333	37	1
4	Old	4	0.753444	29.9138	18.1350	338	0	Stable_NoNotch	1	1	0	0.666667	27	338
5	Old	5	0.804556	29.8687	17.5711	75	0	Unknown	1	1	0	0.666667	23	75
6	Old	6	0.724444	30.0689	17.5922	104	0	Unknown	1	1	0	0.666667	23	104
7	Old	7	0.873222	29.8900	17.5733	3	0	Unknown	0	1	0	0.333333	42	3
8	Old	8	0.673333	30.2856	17.7711	1	0	Unknown	1	1	0	0.666667	36	1
9	Old	9	0.707000	29.9511	17.8738	1	0	Unknown	1	1	0	0.666667	27	1
10	Old	10	0.753111	29.9133	17.6044	130	0	Stable_NoNotch	1	1	0	0.666667	24	130
11	New	1	0.708000	29.9500	16.3900	28	0	FM3_FirstLayer	1	1	0	0.666667		
12	New	2	0.701000	29.9400	16.5000	15	0	FM3_FirstLayer	1	1	0	0.666667		
13	New	3	0.723000	29.8900	16.3300	37	0	FM3_FirstLayer	1	1	0	0.666667		
14	New	4	0.691000	29.8900	16.1000	27	0	FM3_FirstLayer	1	1	0	0.666667		
15	New	5	0.708000	29.9000	16.3600	23	0	FM3_FirstLayer	1	1	0	0.666667		
16	New	6	0.698000	29.9800	16.3800	23	0	FM3_FirstLayer	1	1	0	0.666667		
17	New	7	0.719000	29.9000	16.2700	42	0	FM2_Seam	1	1	0	0.666667		
18	New	8	0.694000	29.9400	16.2700	36	0	FM3_FirstLayer	1	1	0	0.666667		
19	New	9	0.706000	29.9400	16.3200	27	0	FM3_FirstLayer	1	1	0	0.666667		
20	New	10	0.713000	29.9100	16.2400	24	0	FM3_FirstLayer	1	1	0	0.666667		

Figure 37: Old and New hinge life and property table on a part-by-part basis

Parametric lifetime modelling of new batches (following Q3's Weibull) + 4 classes of function plots

Norm	Estimated value	95% CI
Shape β	4.062	[0.988, 6.543]
Scale η	31.094	[26.460, 36.541]
MTTF	28.209	[23.746, 33.511]
t10 (10 per cent time to failure)	17.868	[12.572, 23.022]
AD	1.608	-

Distribution Analysis: Cycles_New

Variable: Cycles_New

Censoring

Censoring Information Count	
Uncensored value	10

Estimation Method: Maximum Likelihood

Distribution: Weibull

Parameter Estimates

Parameter	Estimate	Standard 95.0% Normal CI		
		Error	Lower	Upper
Shape	4.06204	0.987848	2.52197	6.54257
Scale	31.0944	2.56075	26.4595	36.5411

Log-Likelihood = -34.509

Goodness-of-Fit

Anderson-Darling (Adjusted)
1.608

Characteristics of Distribution

	Estimate	Standard 95.0% Normal CI		
		Error	Lower	Upper
Mean(MTTF)	28.2088	2.47876	23.7458	33.5105
Standard Deviation	7.80438	1.52566	5.32034	11.4482
Median	28.4116	2.61056	23.7293	34.0179
First Quartile(Q1)	22.8811	2.92604	17.8084	29.3987
Third Quartile(Q3)	33.6980	2.63490	28.9099	39.2791
Interquartile Range(IQR)	10.8169	2.23946	7.20904	16.2305

Table of Percentiles

Percent	Percentile	Standard 95.0% Normal CI		
		Error	Lower	Upper
1	10.0197	3.12749	5.43460	18.4733
2	11.8988	3.23474	6.98398	20.2725
3	13.1644	3.27006	8.09024	21.4211
4	14.1485	3.27988	8.98225	22.2861
5	14.9666	3.27754	9.74356	22.9894
6	15.6739	3.26851	10.4153	23.5874
7	16.3013	3.25552	11.0212	24.1110
8	16.8682	3.24006	11.5763	24.5792
9	17.3875	3.22301	12.0908	25.0046
10	17.8683	3.20496	12.5721	25.3956
20	21.4939	3.01458	16.3280	28.2943
30	24.1246	2.84503	19.1459	30.3978
40	26.3551	2.70831	21.5473	32.2356
50	28.4116	2.61056	23.7293	34.0179
60	30.4323	2.56261	25.8023	35.8932
70	32.5483	2.58520	27.8561	38.0309
80	34.9592	2.72215	30.0111	40.7231
90	38.1815	3.10017	32.5641	44.7679
91	38.6044	3.16544	32.8731	45.3348
92	39.0609	3.23968	33.2005	45.9558
93	39.5594	3.32508	33.5509	46.6440
94	40.1120	3.42485	33.9311	47.4190
95	40.7369	3.54384	34.3510	48.3100
96	41.4638	3.69012	34.8270	49.3654
97	42.3469	3.87853	35.3884	50.6737
98	43.5031	4.14165	36.0980	52.4273
99	45.2857	4.58023	37.1424	55.2144

Figure 38: New Batch Life Distribution (Weibull) - Parameter Estimation and Life Statistics

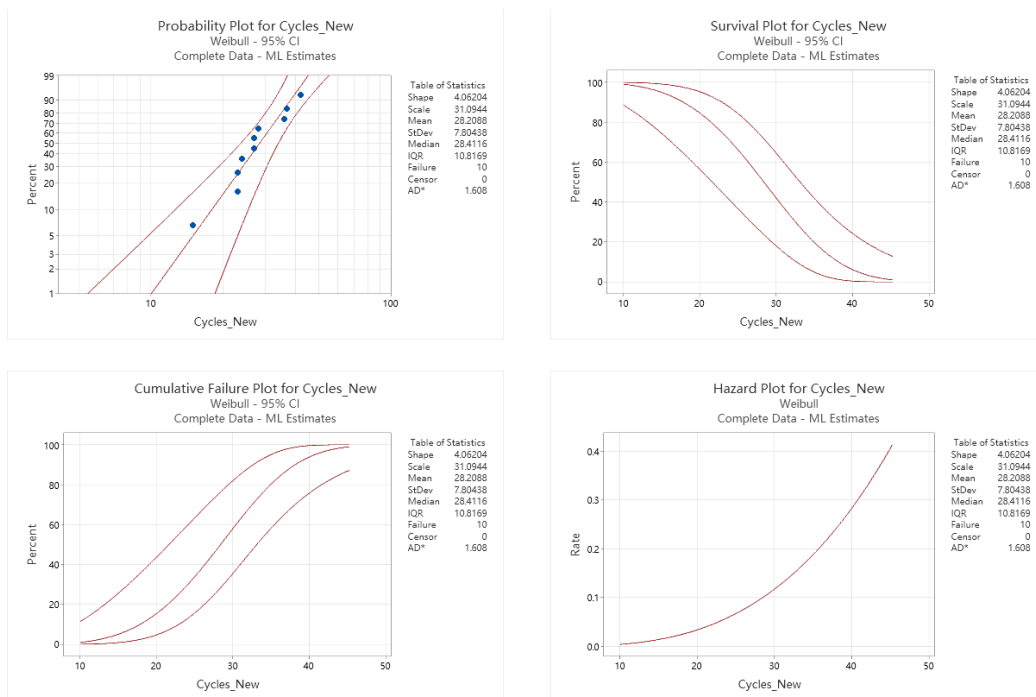


Figure 39: New Batch Life Distribution (Weibull) Four Plots: Probability Plot - Survival - Cumulative Failure - Risk Ratio

Interval estimation: MTTF vs. 10% failure time t_{10}

Norm	Point estimate	95% CI lower limit	95% CI upper limit
MTTF	28.21	23.75	33.51
10% Failure time t_{10} (cycles)	17.87	12.57	23.23

Old vs New life expectancy (Weibull)

Norm	New	Old	Comparison / Interpretation
Weibull geometry β	4.06	0.55	New: risk increases over time (wear and tear dominant); Old: risk decreases over time ("early deaths +

			survivor" structure)
Weibull scale η	31.09	64.69	Old features last longer
B10 (10 per cent time to failure)	17.9 cycles ($\approx[12.6, 23.3]$)	1.08 cycles ($\approx[0.16, 6.10]$)	Early reliability: 16-17 $\times$ for New $\approx$ Old
Median life expectancy (Median)	28.41 cycles	33.20 cycles	Median New $\approx 0.86\times$ of Old (slightly shorter)
MTTF	28.21 ([23.75, 33.51])	110.26 ([32.90, 369.46])	Mean lifetime Old $\approx 3.9\times$ of New (Old interval is very wide, uncertainty is high with small samples)
R (30)	≈ 0.42	≈ 0.52	Old has a better survival rate at 30.
R (50)	≈ 0.001	≈ 0.42	At 50, Old is still $\sim 42\%$ alive, New is almost depleted.

Distribution Analysis: Cycles by Batch

Variable: Cycles
Batch = New

Censoring

Censoring Information Count	
Uncensored value	10

Estimation Method: Maximum Likelihood

Distribution: Weibull

Parameter Estimates

Parameter	Estimate	Error	Standard 95.0% Normal CI	
			Lower	Upper
Shape	4.06204	0.987848	2.52197	6.54257
Scale	31.0944	2.56075	26.4595	36.5411

Log-Likelihood = -34.509

Goodness-of-Fit

Anderson-Darling (Adjusted)	
1.608	

Characteristics of Distribution

	Estimate	Error	Standard 95.0% Normal CI	
			Lower	Upper
Mean(MTTF)	28.2088	2.47876	23.7458	33.5105
Standard Deviation	7.80438	1.52566	5.32034	11.4482
Median	28.4116	2.61056	23.7293	34.0179
First Quartile(Q1)	22.8811	2.92604	17.8084	29.3987
Third Quartile(Q3)	33.6980	2.63490	28.9099	39.2791
Interquartile Range(IQR)	10.8169	2.23946	7.20904	16.2305

Table of Percentiles

Percent	Percentile	Error	Standard 95.0% Normal CI	
			Lower	Upper
1	10.0197	3.12749	5.43460	18.4733
2	11.8988	3.23474	6.98398	20.2725
3	13.1644	3.27006	8.09024	21.4211
4	14.1485	3.27988	8.98225	22.2861
5	14.9666	3.27754	9.74356	22.9894
6	15.6739	3.26851	10.4153	23.5874
7	16.3013	3.25552	11.0212	24.1110
8	16.8682	3.24006	11.5763	24.5792
9	17.3875	3.22301	12.0908	25.0046
10	17.8683	3.20496	12.5721	25.3956
20	21.4939	3.01458	16.3280	28.2943
30	24.1246	2.84503	19.1459	30.3978
40	26.3551	2.70831	21.5473	32.2356
50	28.4116	2.61056	23.7293	34.0179
60	30.4323	2.56261	25.8023	35.8932
70	32.5483	2.58520	27.8561	38.0309
80	34.9592	2.72215	30.0111	40.7231
90	38.1815	3.10017	32.5641	44.7679
91	38.6044	3.16544	32.8731	45.3348
92	39.0609	3.23968	33.2005	45.9558
93	39.5594	3.32508	33.5509	46.6440
94	40.1120	3.42485	33.9311	47.4190
95	40.7369	3.54384	34.3510	48.3100
96	41.4638	3.69012	34.8270	49.3654
97	42.3469	3.87853	35.3884	50.6737
98	43.5031	4.14165	36.0980	52.4273
99	45.2857	4.58023	37.1424	55.2144

Figure 40: New Batch Life Distribution (Weibull) Parameters and Percentile Tables

Distribution Analysis: Cycles by Batch

Variable: Cycles
Batch = Old

Censoring

Censoring Information Count	
Uncensored value	10

Estimation Method: Maximum Likelihood

Distribution: Weibull

Parameter Estimates

Parameter	Estimate	Standard Error	95.0% Normal CI	
			Lower	Upper
Shape	0.549572	0.146079	0.326415	0.925292
Scale	64.6858	39.1563	19.7493	211.869

Log-Likelihood = -52.671

Goodness-of-Fit

Anderson-Darling (Adjusted)
1.922

Characteristics of Distribution

	Estimate	Standard Error	95.0% Normal CI	
			Lower	Upper
Mean(MTTF)	110.256	68.0244	32.9029	369.459
Standard Deviation	216.876	175.959	44.2180	1063.71
Median	33.2028	22.6289	8.73075	126.270
First Quartile(Q1)	6.70266	6.55380	0.986165	45.5559
Third Quartile(Q3)	117.200	67.5295	37.8858	362.561
Interquartile Range(IQR)	110.498	63.6063	35.7579	341.456

Table of Percentiles

Percent	Percentile	Standard Error	95.0% Normal CI	
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4	0.191952	0.350870	0.0053368	6.90406
5	0.290828	0.501221	0.0099232	8.52353
6	0.409125	0.670227	0.0164976	10.1459
7	0.546829	0.856459	0.0253907	11.7768
8	0.704027	1.05883	0.0369329	13.4204
9	0.880888	1.27650	0.0514566	15.0800
10	1.07764	1.50879	0.0692978	16.7583
20	4.22176	4.55763	0.508830	35.0279
30	9.91116	8.87862	1.71238	57.3652
40	19.0538	14.6864	4.20617	86.3133
50	33.2028	22.6289	8.73075	126.270
60	55.1728	34.1941	16.3750	185.896
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93	383.445	246.231	108.918	1349.92
94	424.848	278.152	117.745	1532.93
95	476.270	318.972	128.168	1769.82
96	542.784	373.521	140.885	2091.17
97	634.270	451.416	157.205	2559.07
98	774.002	575.924	180.042	3327.44
99	1041.48	829.486	218.631	4961.23

Figure 41: Old Batch Life Distribution (Weibull) Parameters and Percentile Tables

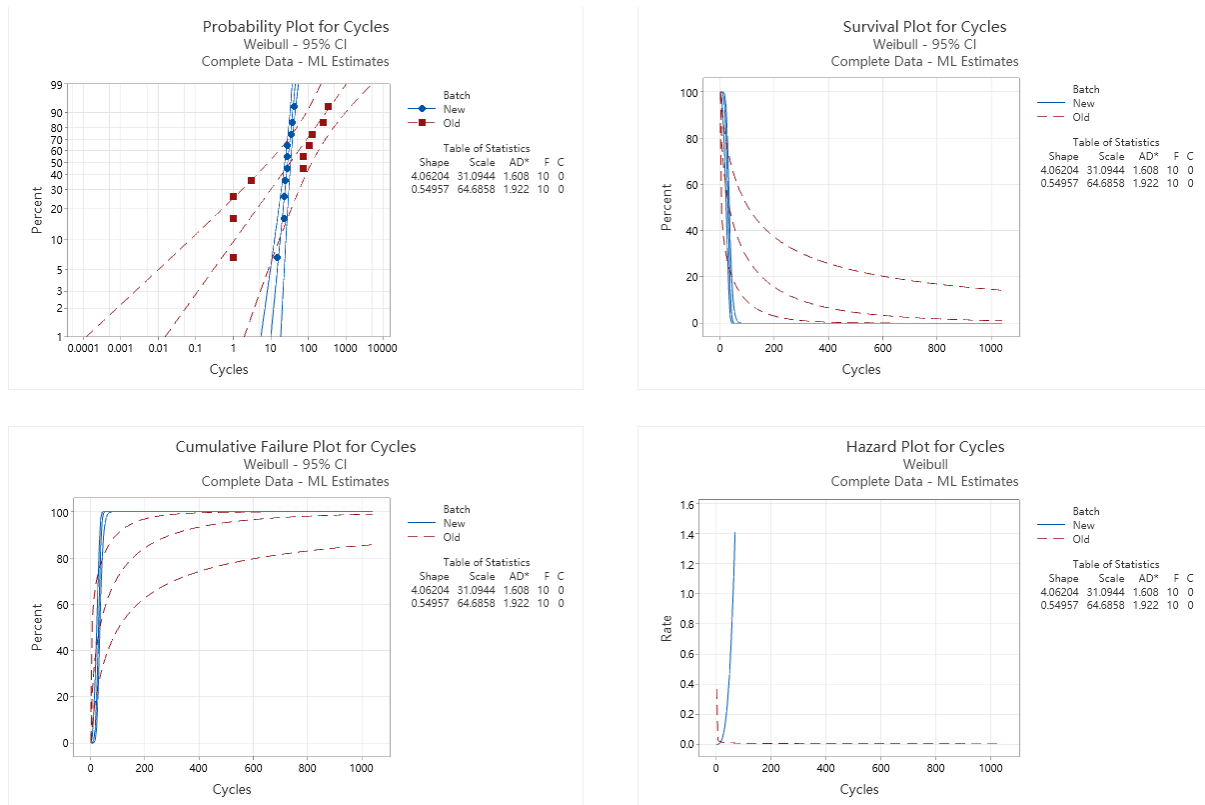


Figure 42: Old vs New: Weibull comparison (probability/survival/cumulative failure/risk rate)